

Computer Vision 1

2-1: In Place Processing, Photometric Stereo

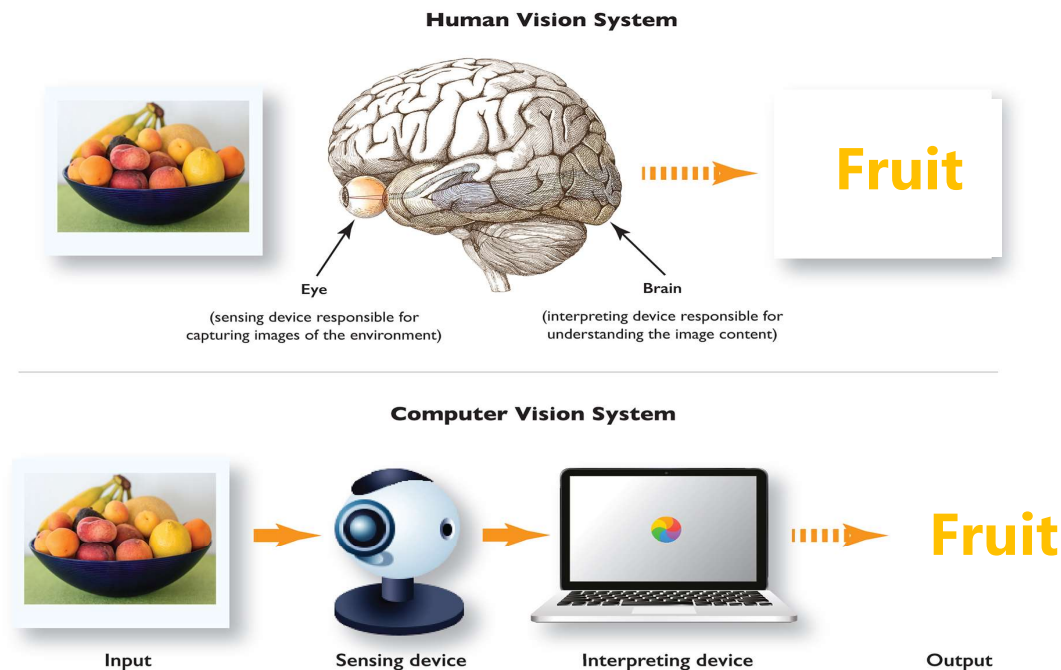
Master of AI – 2024/2025
University of Amsterdam

Arnoud Visser & Martin Oswald

Based on a presentation of Shaodi You,
augmented with material of Joe Redmons & many others

The computer vision problem

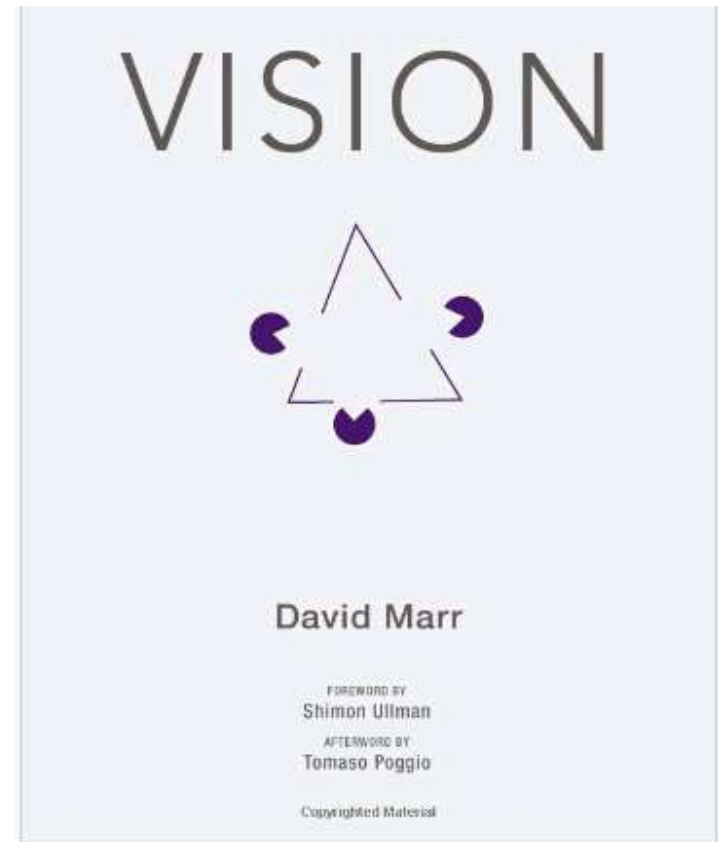
- Make a computer see and understand images



- We know it is physically possible –
we do it every day and effortlessly!

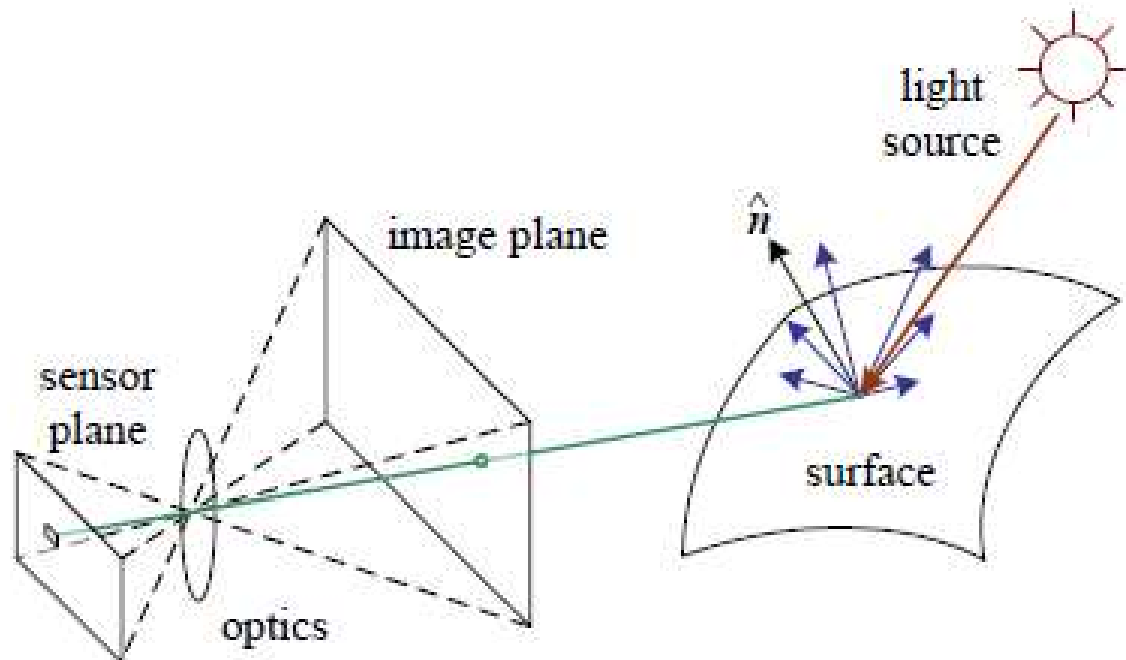
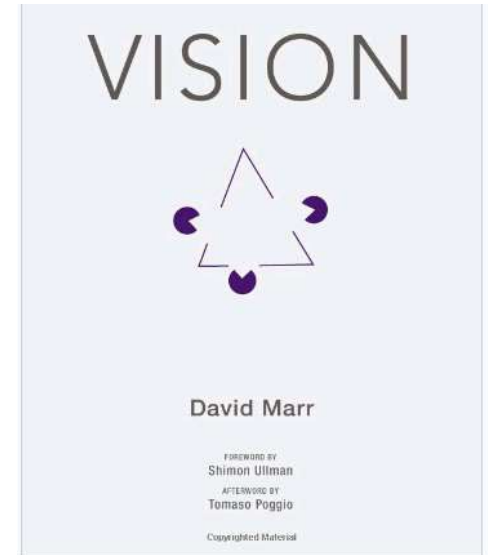
Three levels of vision processing

- Low-level vision:
 - image processing, denoising, filtering, image restoration, low-level feature extraction.
- Mid-level vision:
 - image analysis, image segmentation, contour extraction, perceptual organization, 2-1/2D representation, 3D information recovery.
- High-level vision:
 - visual detection, recognition, understanding, semantic labelling, activity, event detection.

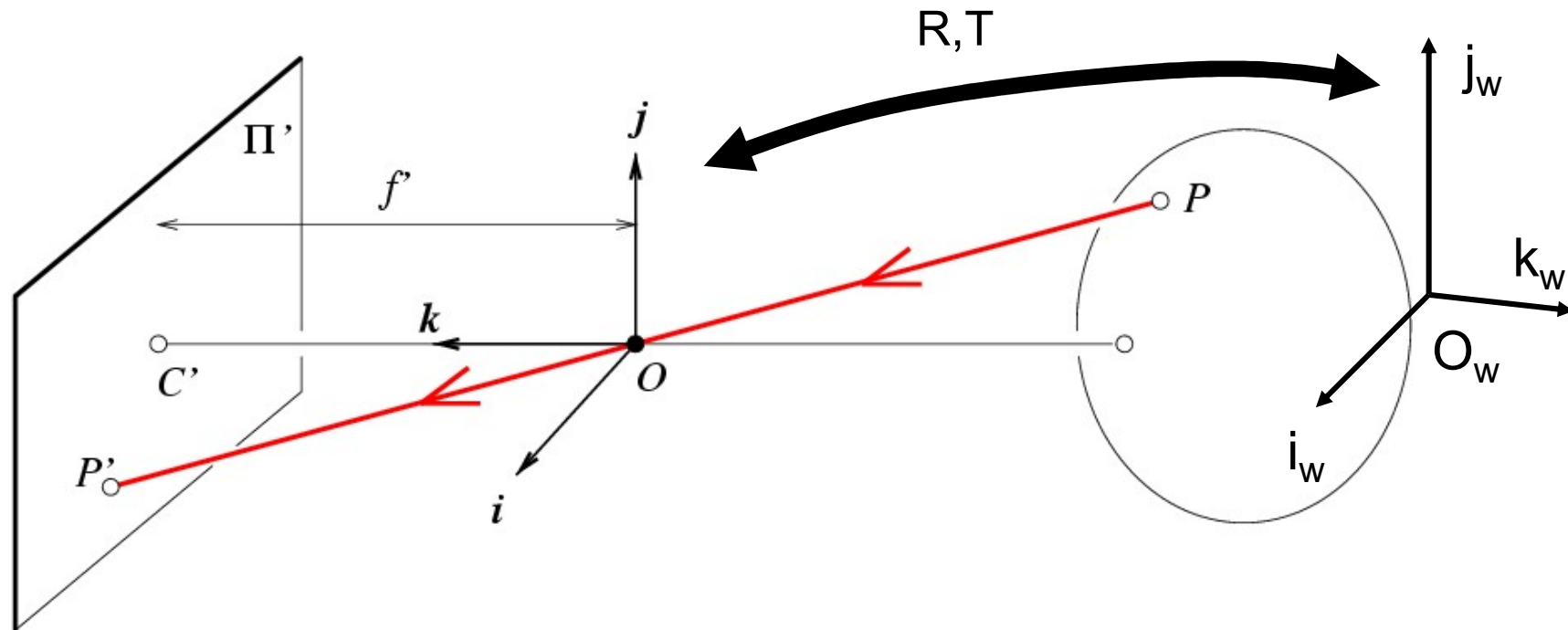


4 main factors which define intensity values

- the geometry of the scene
- the reflectances of the visible surface
- the illumination of the scene
- the viewpoint



Projection matrix

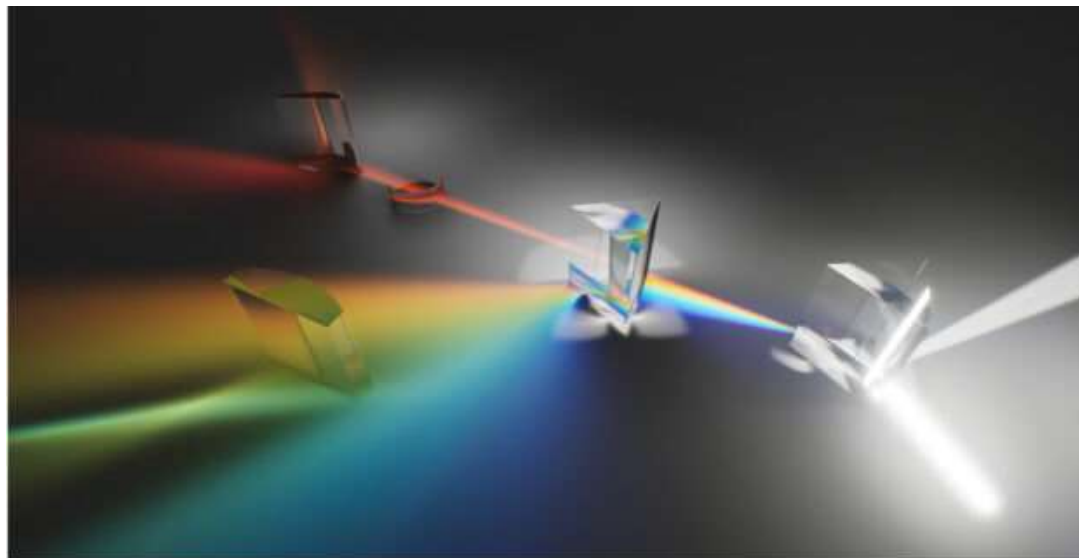


$$\mathbf{x} = \mathbf{K}[\mathbf{R} \quad \mathbf{t}] \mathbf{X}$$

Extrinsic Matrix

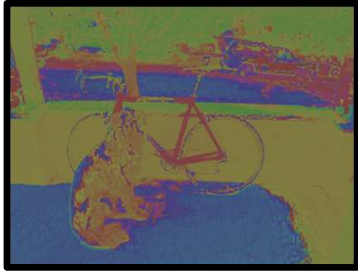
$\mathbf{x}$: Image Coordinates: $(u, v, 1)$
 $\mathbf{K}$: Intrinsic Matrix (3×3)
 $\mathbf{R}$: Rotation (3×3)
 $\mathbf{t}$: Translation (3×1)
 $\mathbf{X}$: World Coordinates: $(X, Y, Z, 1)$

Absorption and reflection



Nvidia, [Generating Ray-Traced Caustic Effects in Unreal Engine 4](#), Dec. 2020

More value = lighter image



2x



Textbook

- 3.1.1
- 3.1.2
- 3.1.4
- 2.2.2
- 13.1.1

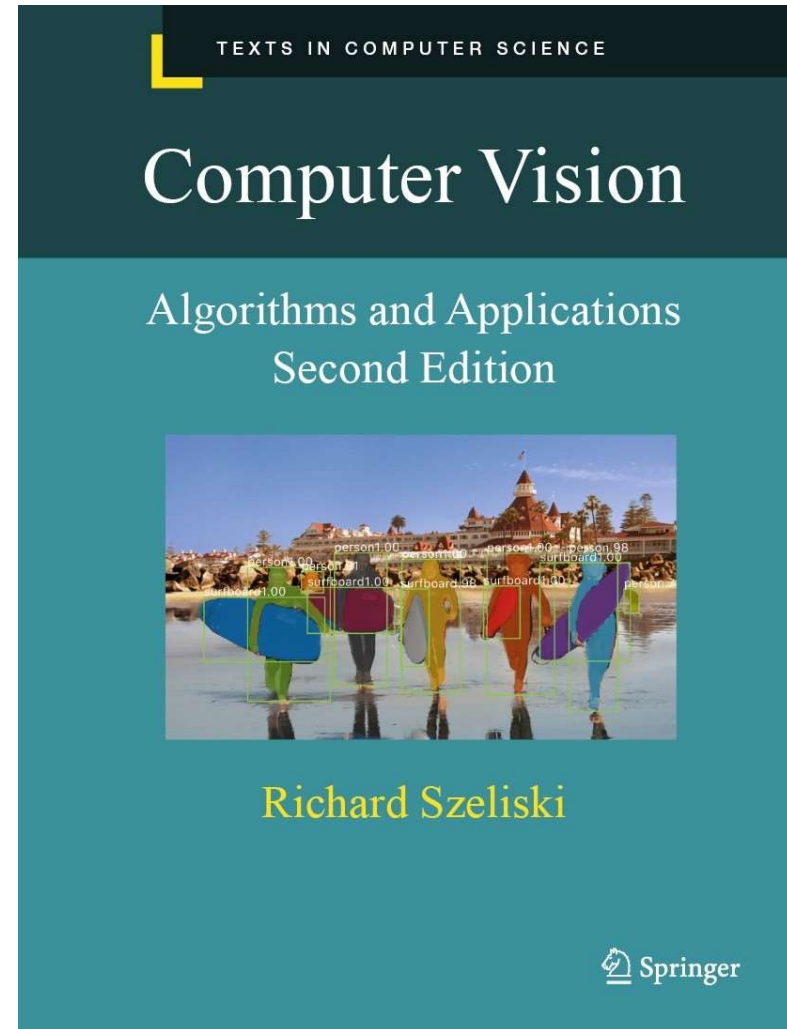
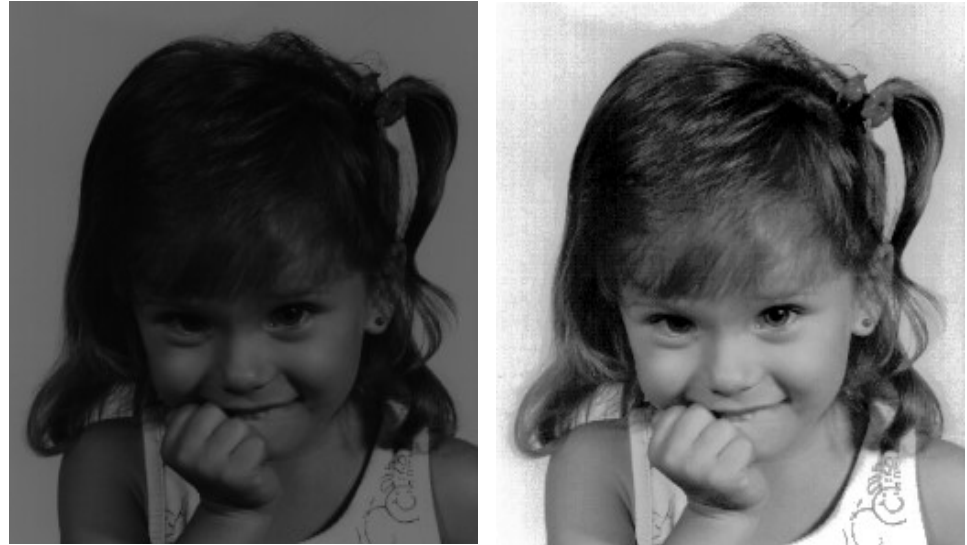


Image processing operations

- Point-wise operations
 - histogram operation
- Neighborhood operations
 - Spatial domain (image convolution)
 - Frequency domain (Fourier transform)
 - Binary image analysis
- Geometric operations
 - Rotation, translation, affine...

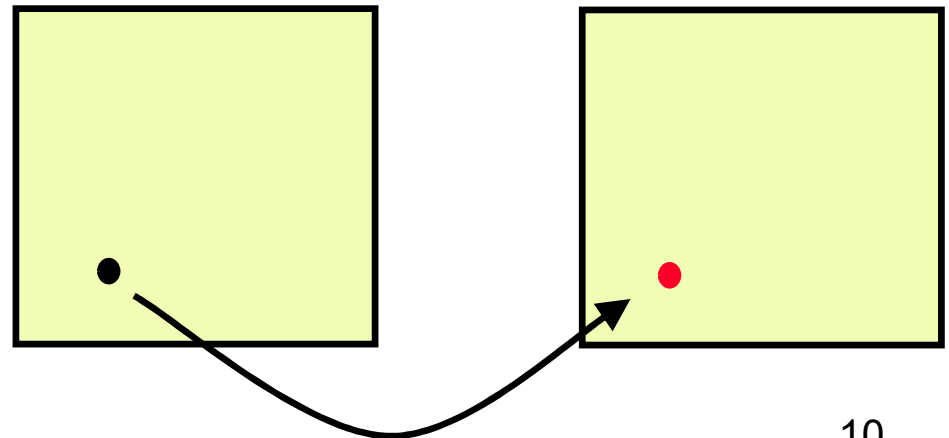
Point Operations

- Operation on ***Pixel's value***
- Context free
- Operation can be performed on the image histogram
- Example:



$$g(x, y) = M(f(x, y))$$

$$g(x, y) = \alpha \cdot f(x, y) + \beta$$

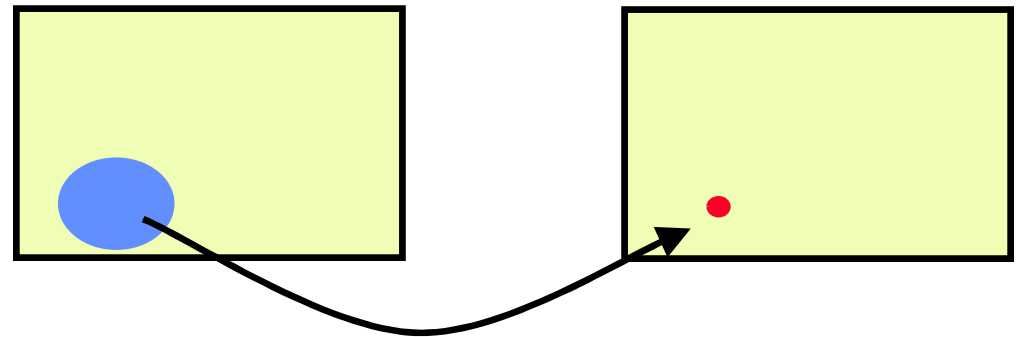


Neighbourhood Operations

- Operation depends on ***Pixel's value*** and its spatial ***neighbours***.
- Context dependant.
- Example:



$$g(x, y) = M(\{f(i, j) | (i, j) \in N(x, y)\})$$



$$g(x, y) = \sum_{i, j \in N(x, y)} f(i, j) / n$$

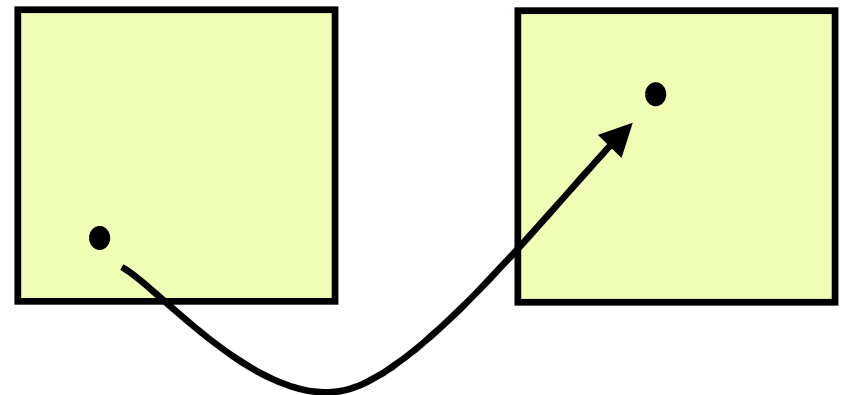
Geometric Operations

- Operation depend on ***pixel's coordinates***.
- Independent of pixels value.
- Context free.
- Example: translation

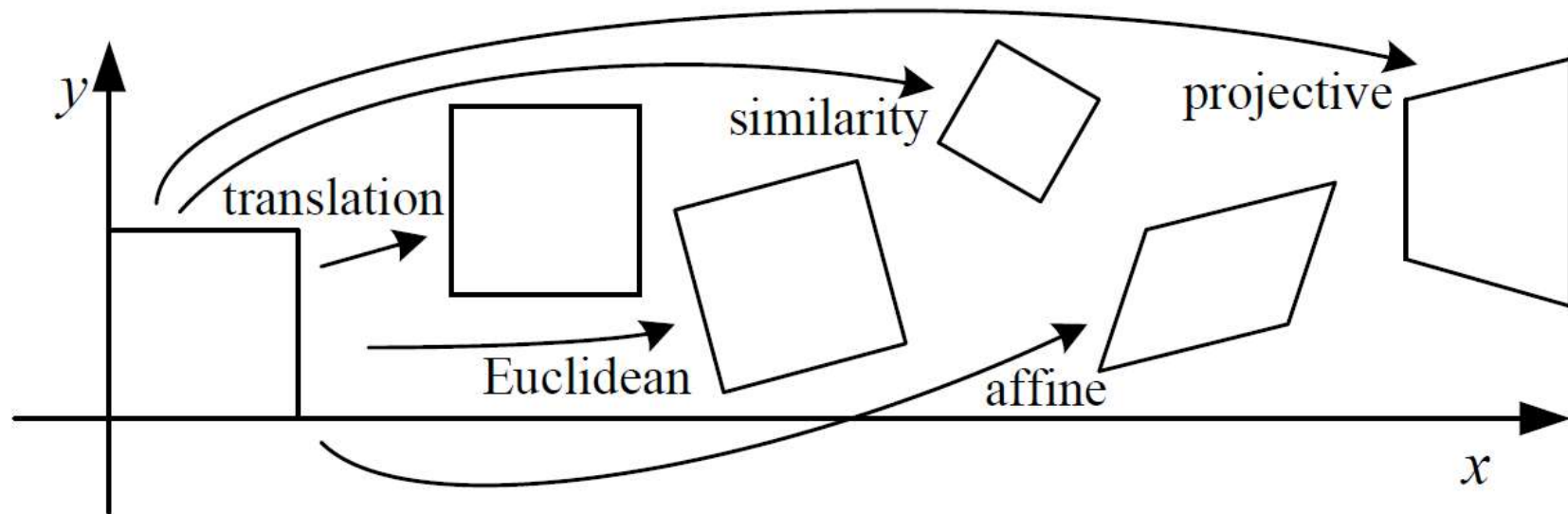
$$g(x, y) = f(x + a, y + b)$$

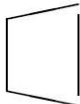


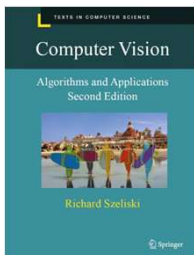
$$g(x, y) = f(G(x, y))$$



Geometric Operations



Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	3	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	6	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	7	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{3 \times 4}$	12	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{4 \times 4}$	15	straight lines	



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- In Place Processing
 - Histogram
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 - Histogram Equalization
- Reflection Model
 - Ideal diffuse
 - Ideal specular
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Histogram

The histogram of a digital image with gray values

$$r_0, r_1, \dots, r_{L-1}$$

is the discrete function

$$p(r_k) = \frac{n_k}{n}$$

n_k : Number of pixels with gray value r_k

n : total Number of pixels in the image

The function $p(r_k)$ represents the fraction of the total number of pixels with gray value r_k .

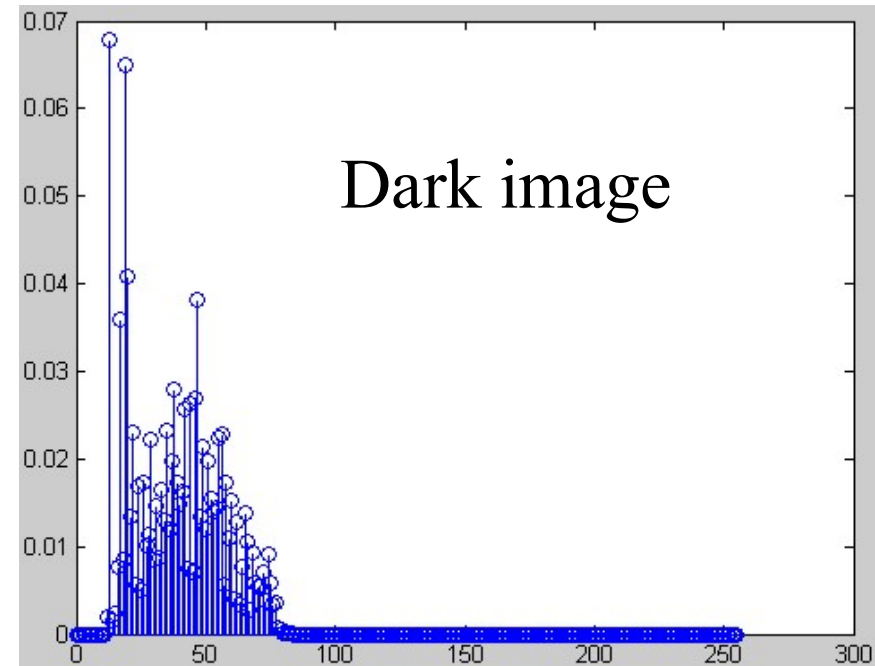
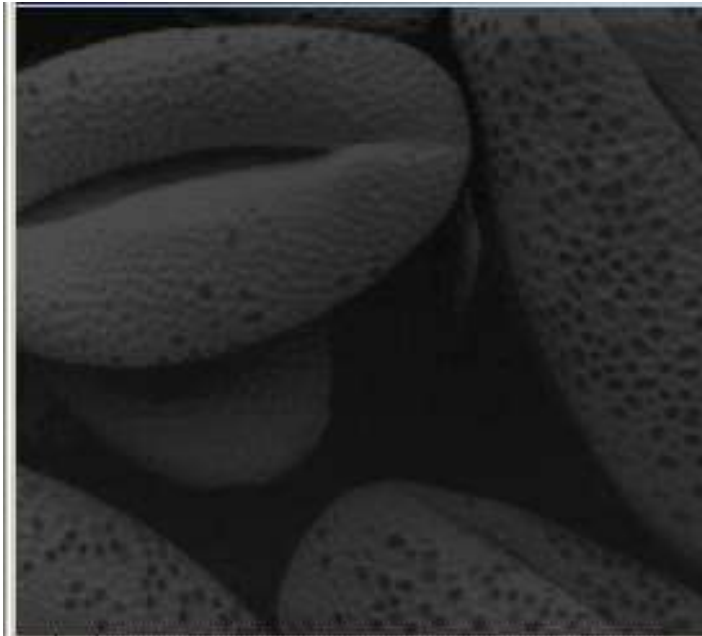
Histogram

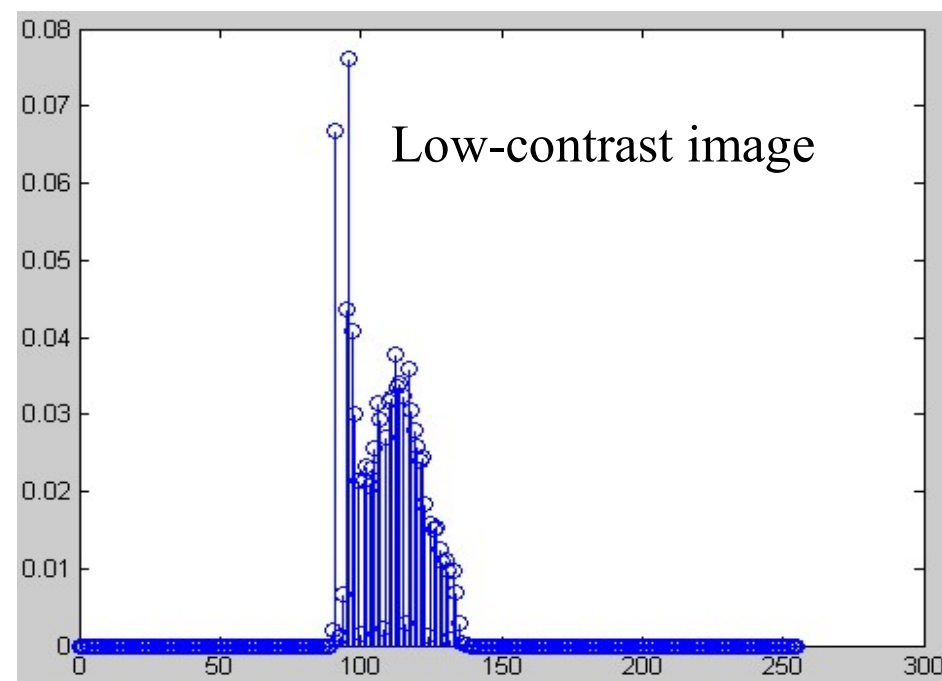
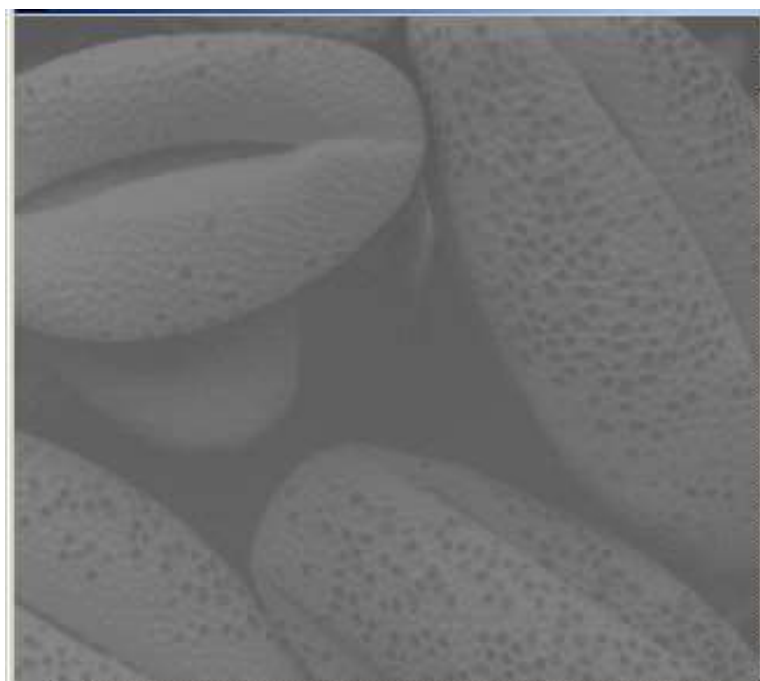
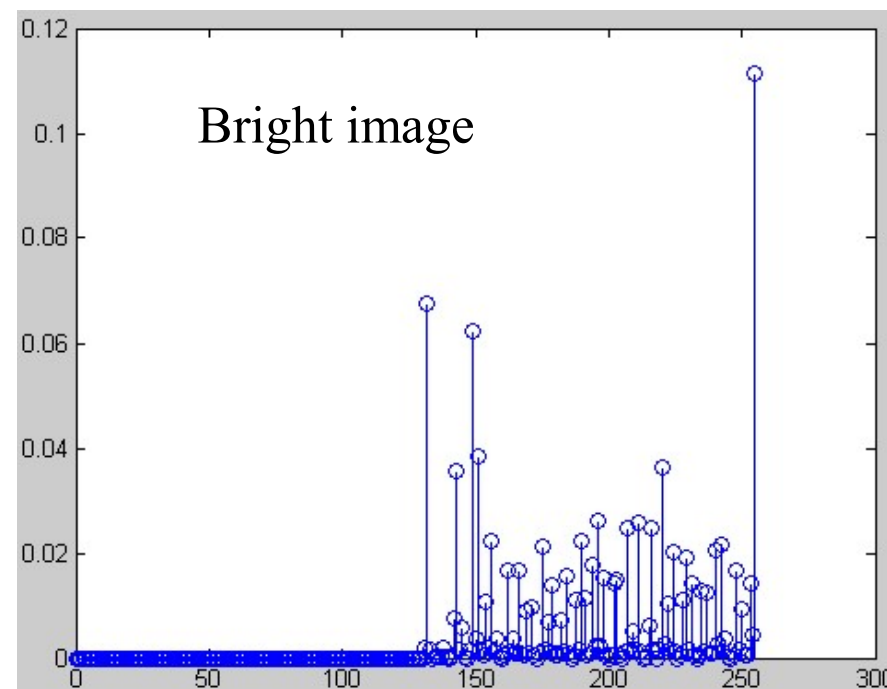
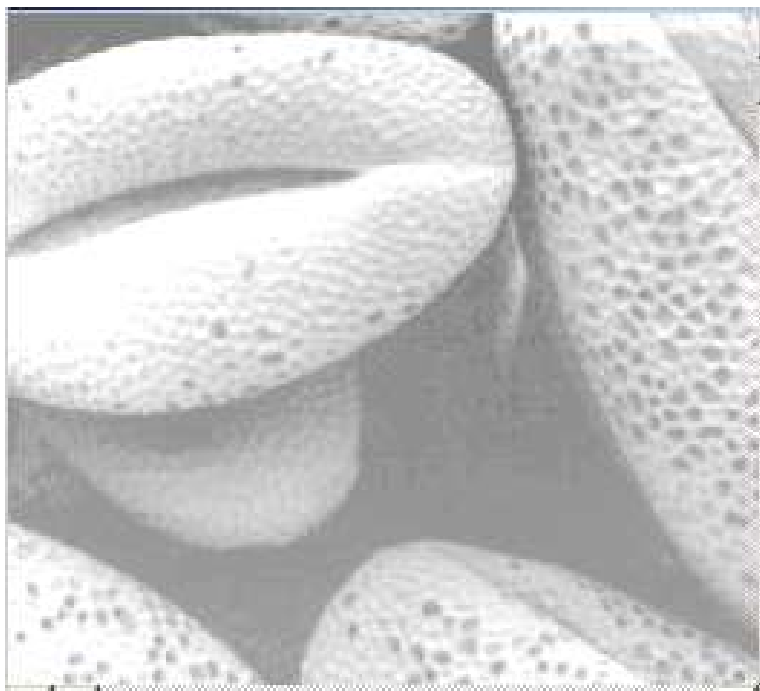
- Histogram provides a global description of the **appearance** of the image.
- If we consider the gray values in the image as realizations of a random variable R , with some probability density, histogram provides an approximation to this **probability density**.
- In other words,

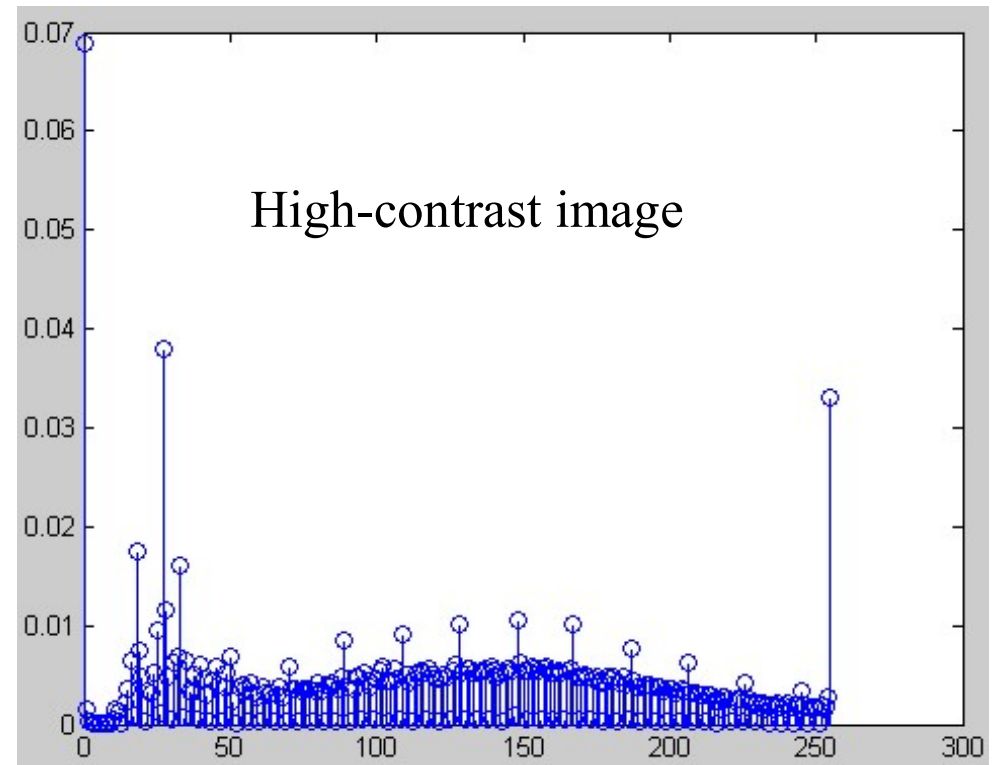
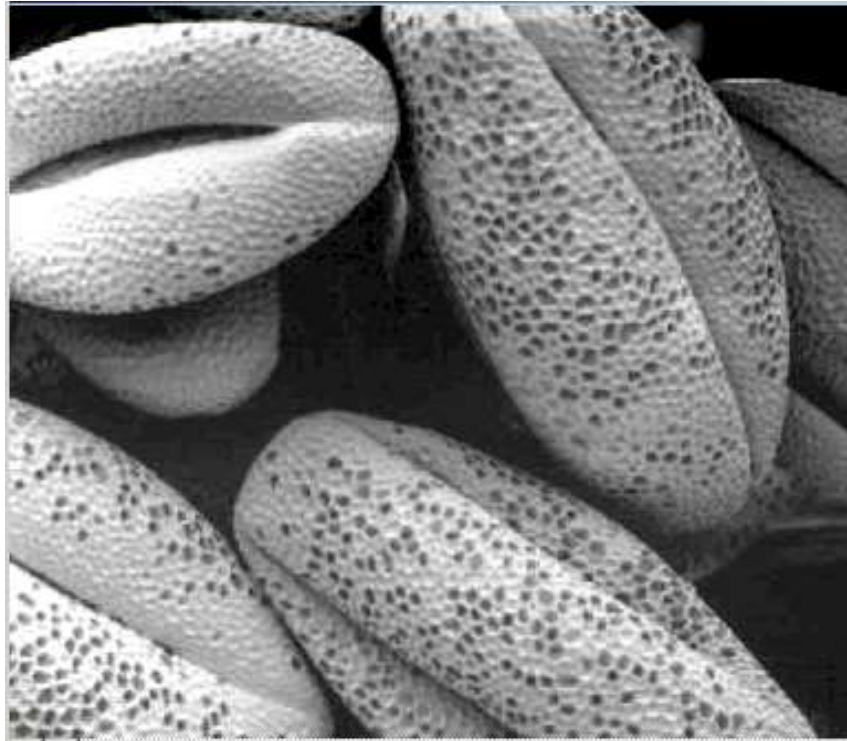
$$\Pr(R = r_k) \approx p(r_k)$$

Some Typical Histograms

- The shape of a histogram provides useful information for image global appearance.



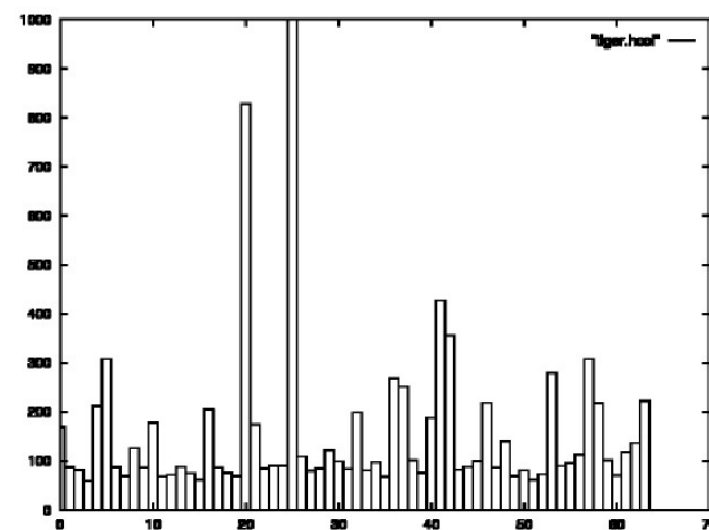
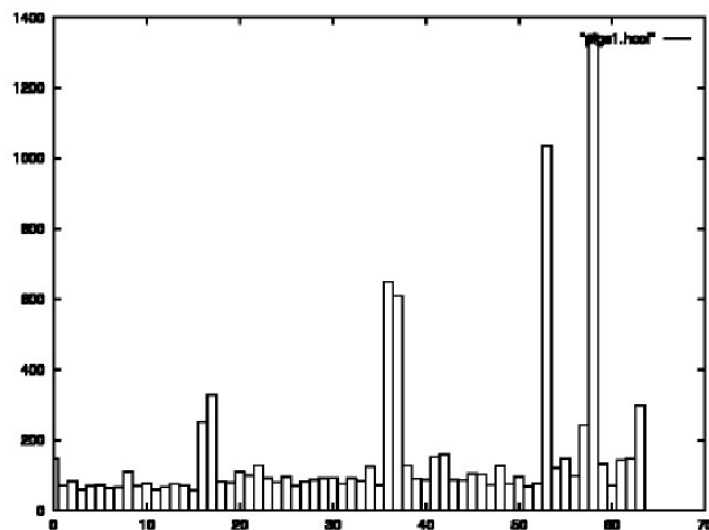
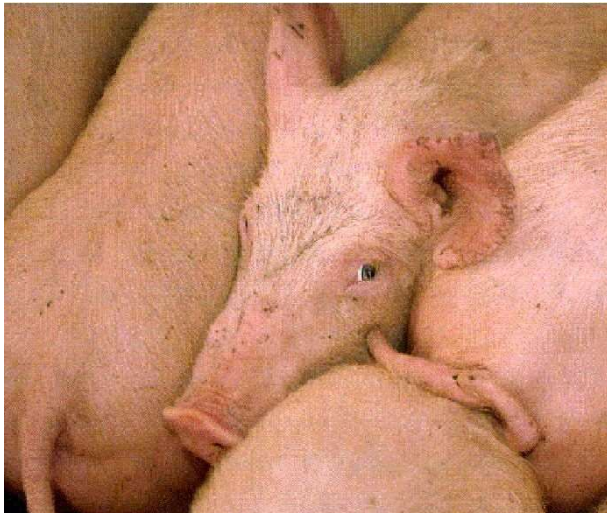




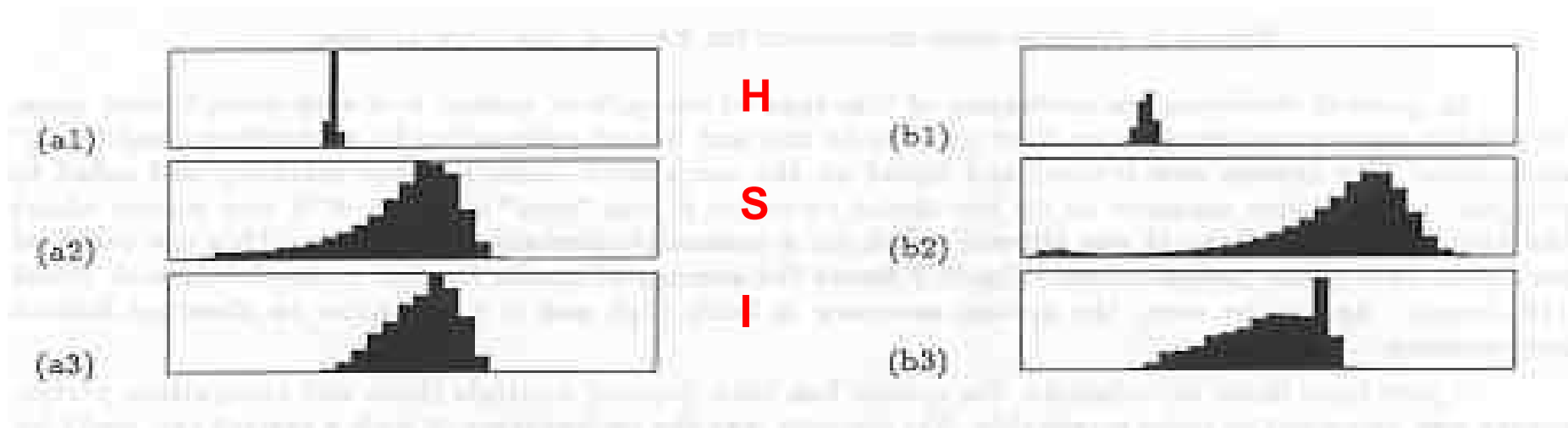
Color histograms can represent an image

- Histogram is fast and easy to compute.
- Size can easily be normalized so that different image histograms can be compared.
- Can match color histograms for database query or classification.

Histograms of two color images



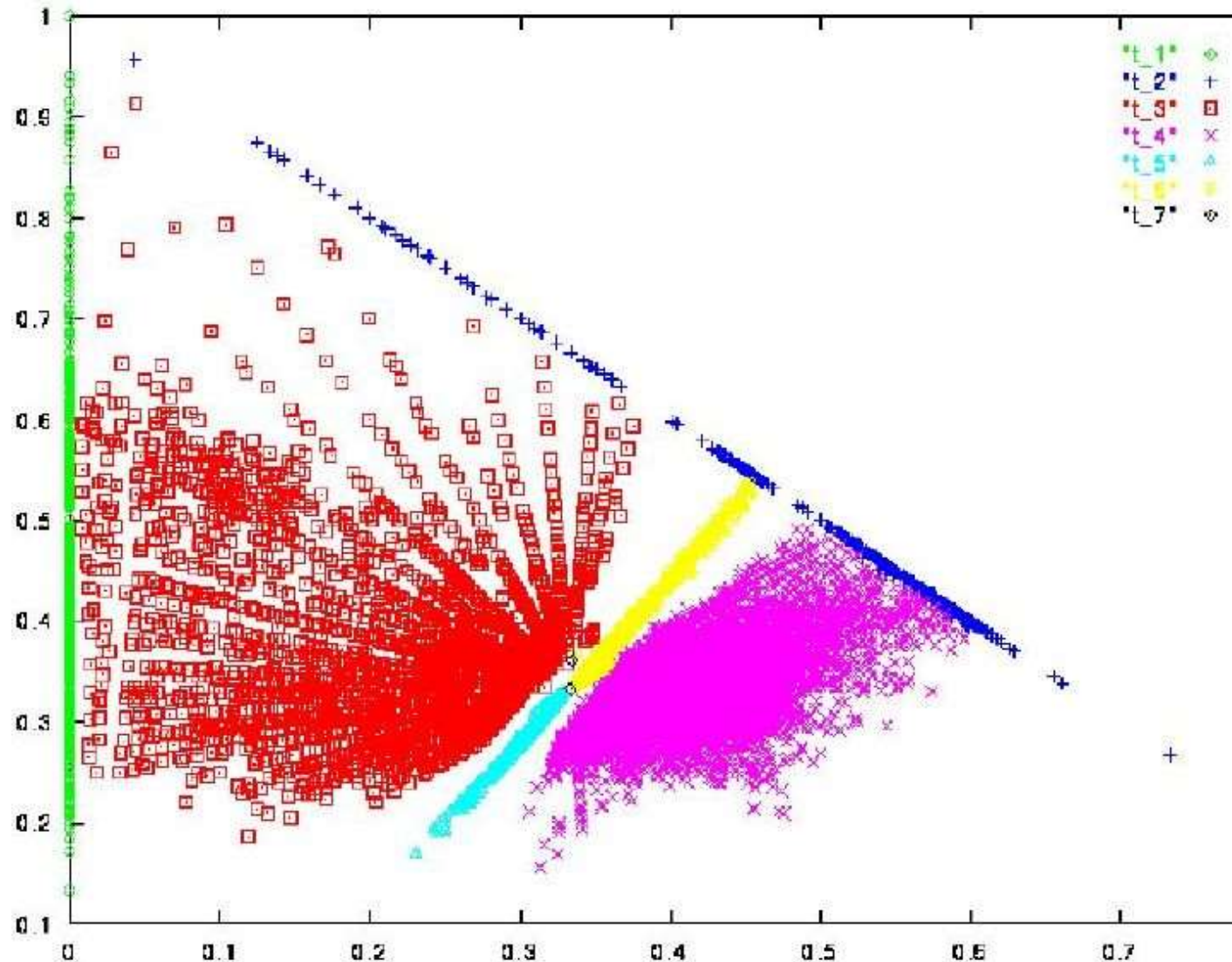
Apples versus Oranges



Separate HSI histograms for apples (left) and oranges (right) used by IBM's VeggieVision for recognizing produce at the grocery store checkout station

Skin color in Normalized R-G Space

G normalized



Purple region shows skin color samples from several people. Blue and yellow regions show skin in shadow or behind a beard.

R normalized 23

Finding a face in video frame



- (left) input video frame
- (center) pixels classified according to RGB space
- (right) largest connected component with aspect similar to a face (all work contributed by Vera Bakic)

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Histogram modification

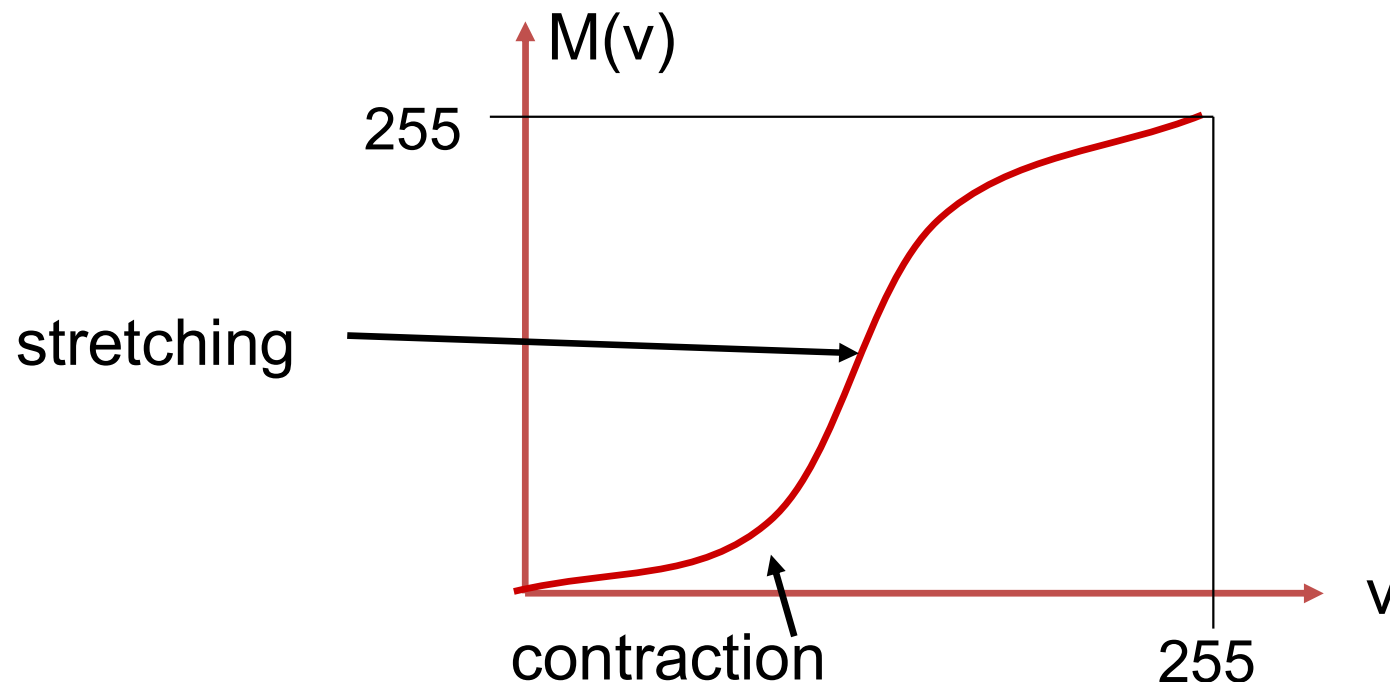
- Given a point operation:

$$v_b = M(v_a)$$

- $M(v_a)$ takes any value v_a in image A and maps it into v_b in image B.
- **Requirement:** the mapping M is a non descending function (M^{-1} exists).
- In this case, the area under H_a between 0 and v_a is equal to the area under H_b between 0 and v_b .

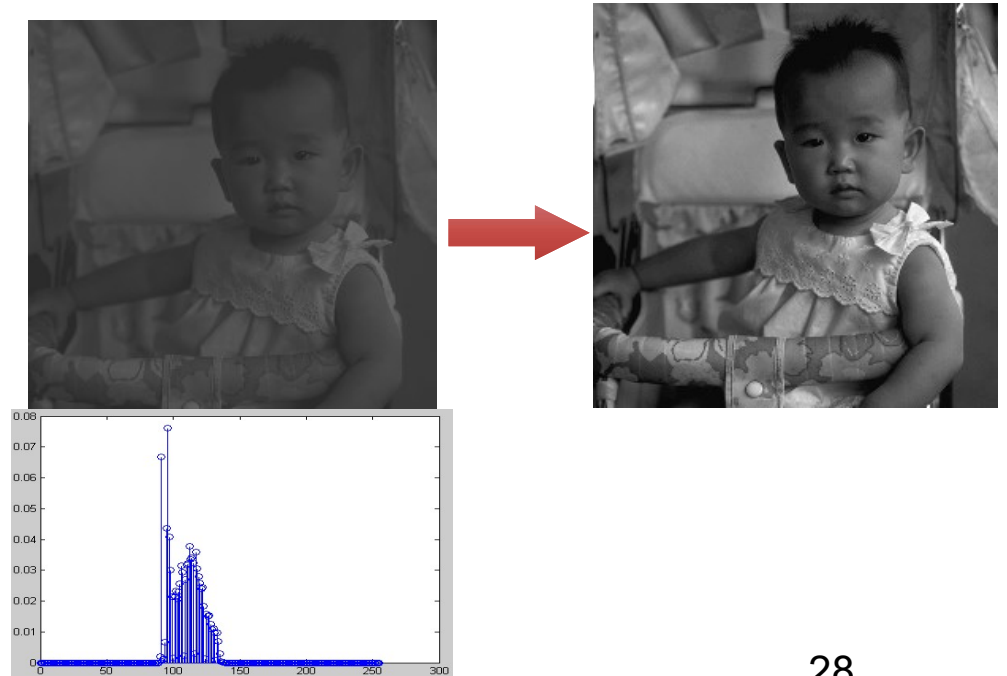
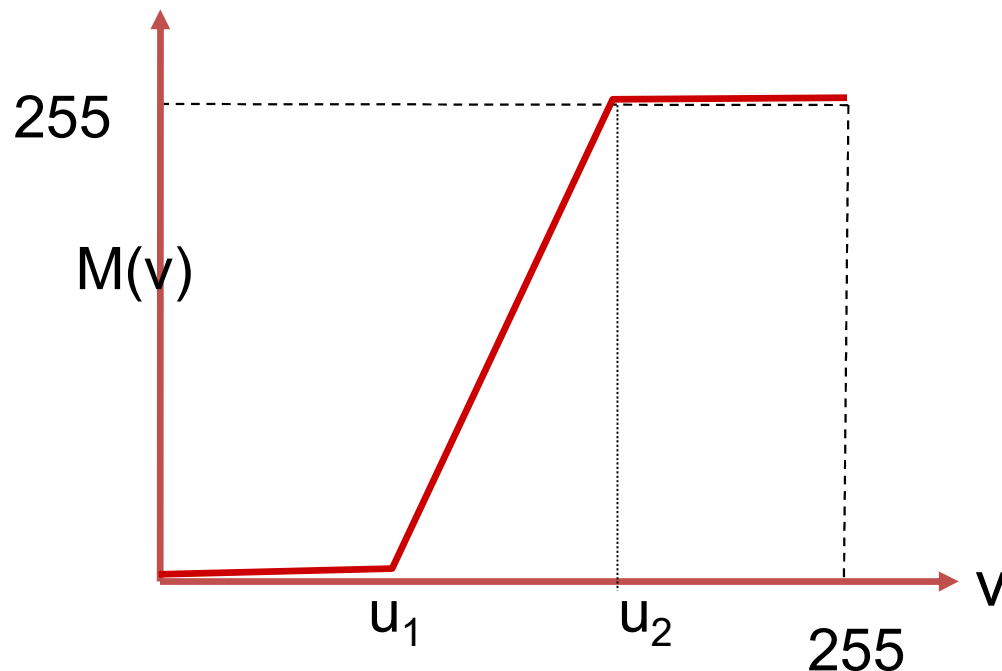
Monotonicity Requirement

- If it is required to map the full gray-level range (256 values) to its full range while keeping the gray-level order, a **non-decreasing monotonic mapping function** is needed:



Contrast Enhancement

- If most of the gray-levels in the image are in $[u_1, u_2]$, the following mapping increases the image contrast.
- The values u_1 and u_2 can be found by using the image's accumulated histogram.



Power-law transformations

$$s = cr^\gamma$$

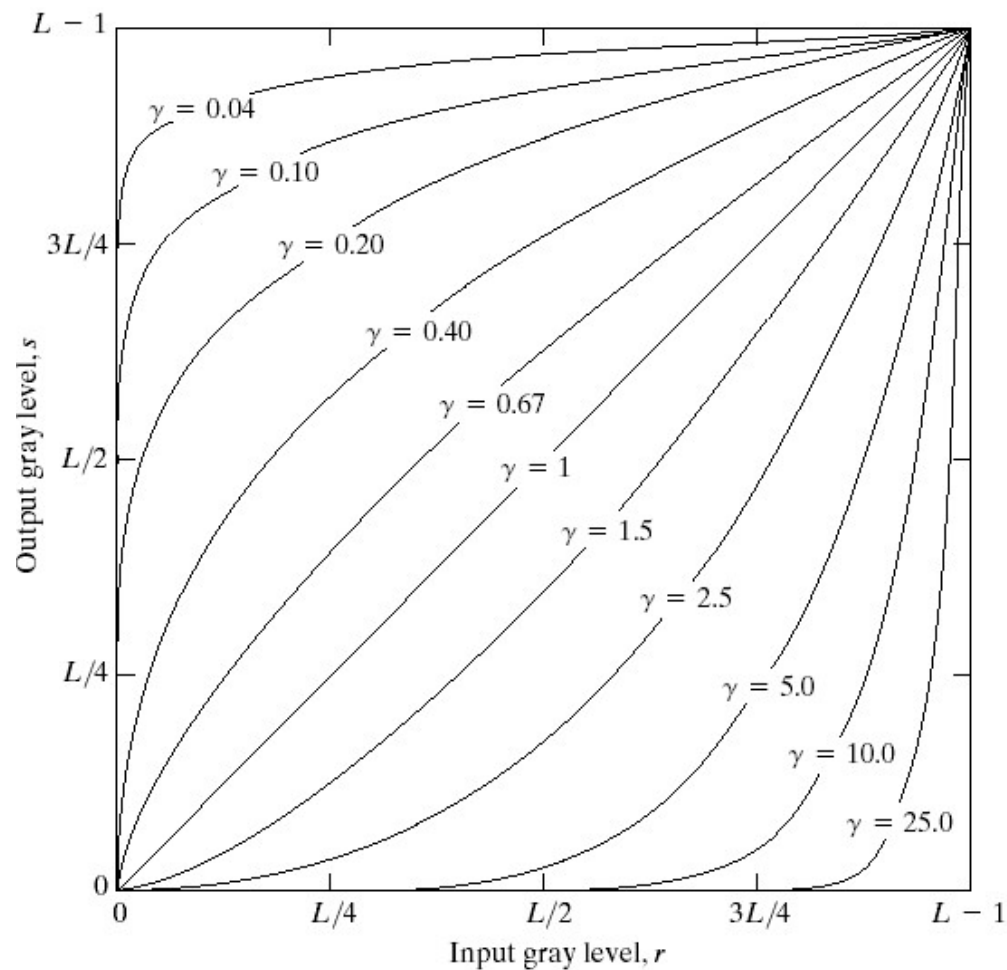


FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases).

a b
c d

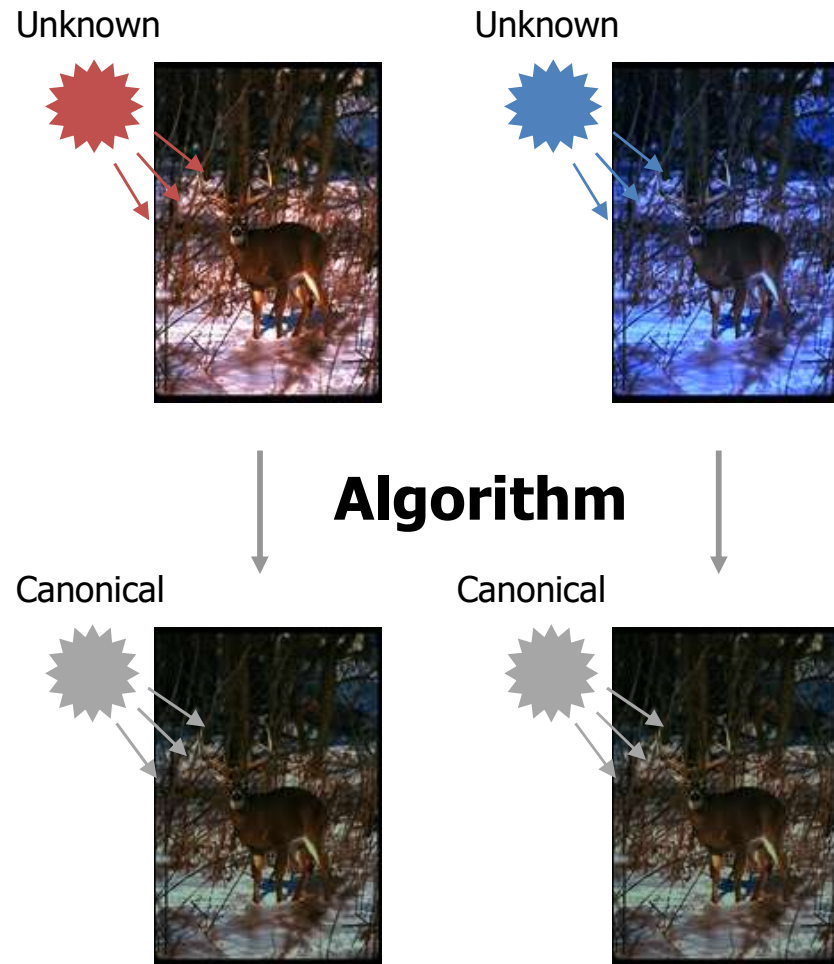
FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of
applying the
transformation in
Eq. (3.2-3) with
 $c = 1$ and
 $\gamma = 3.0, 4.0$, and
 5.0 , respectively.
(Original image
for this example
courtesy of
NASA.)



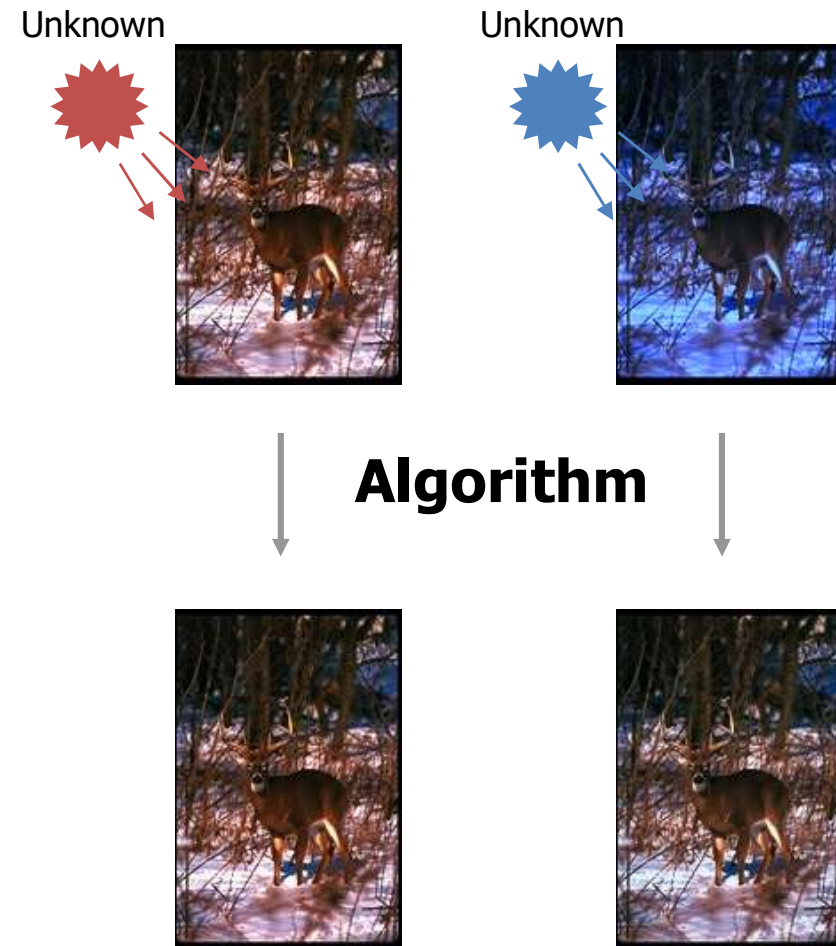
Color constancy

Gray-world



Mean color = constant = $[128, 128, 128]$

Scale-by-max



Max color = constant = 255

Gray world algorithm (slides no. 94): linear stretch the pixel values of the image channel wise, so the averages of each channel are all 128.
Max-RGB algorithm (slides no. 94): linear stretch the pixel values of the image (channel wise), so that the brightest pixel has the value [255 255 255].

https://en.wikipedia.org/wiki/Color_normalization

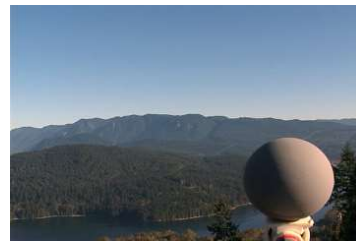
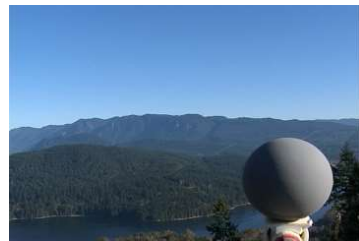
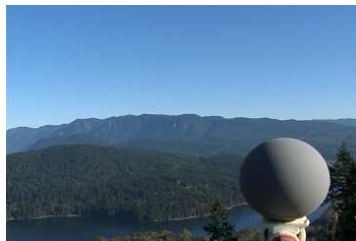
White Patch and Grey-world Examples

Original image

Ideal correction

White-Patch

Grey-world

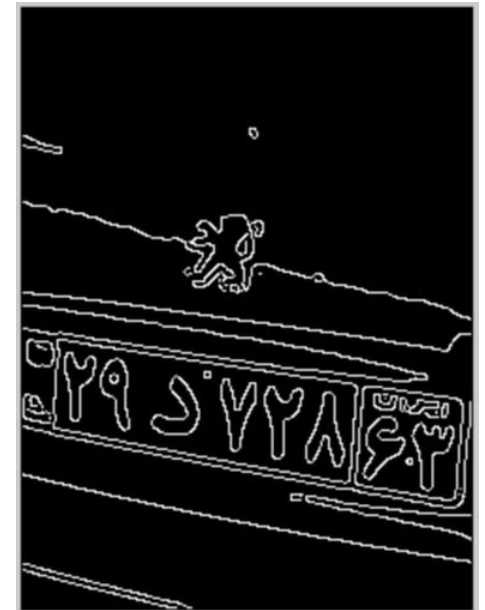
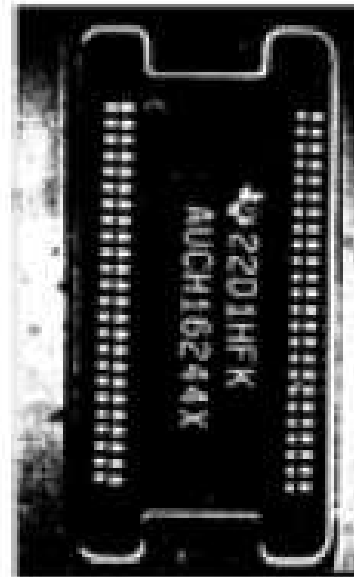
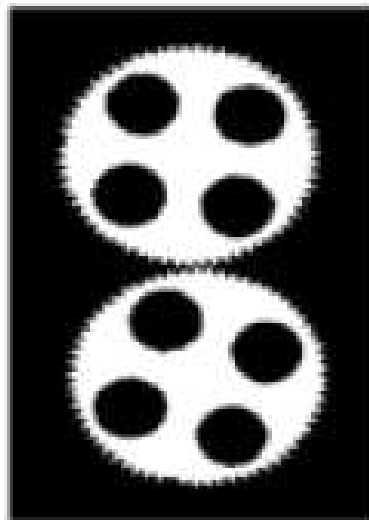
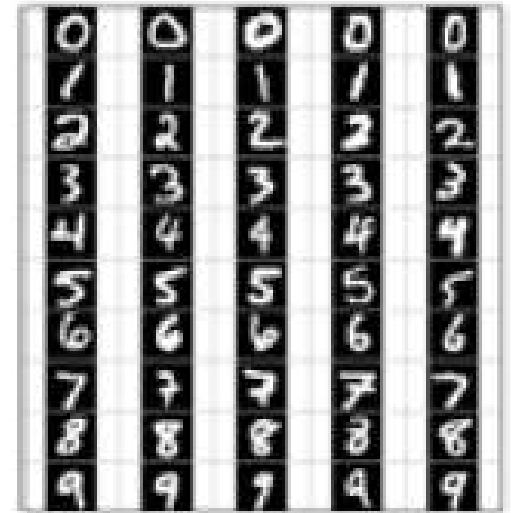


Images from “A large image data set for color constancy research”, by F. Ciurea and B. Funt in CIC 2003.

Image threshold (histogram binarization):

how to get a binary image from a colour one ?

Why binary ?



Thresholding

- Given a grayscale image or an intermediate matrix → threshold to create a binary output.

Example: intensity-based detection



```
fg_pix = find(im < 65);
```

Looking for dark pixels

Thresholding

- Given a grayscale image or an intermediate matrix $\rightarrow$ threshold to create a binary output.

Example: color-based detection



```
fg_pix = find(hue > t1 & hue < t2);
```

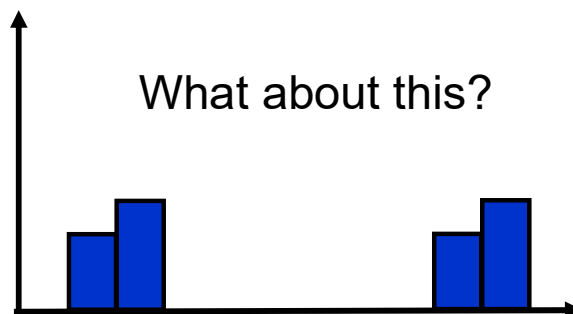
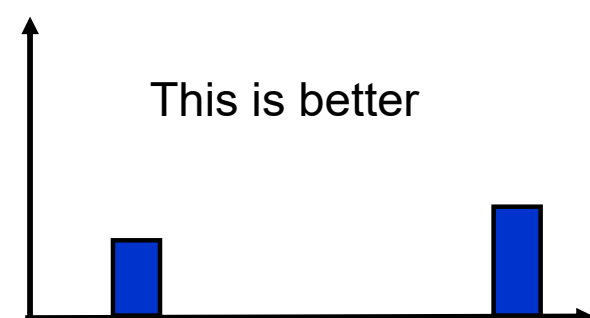
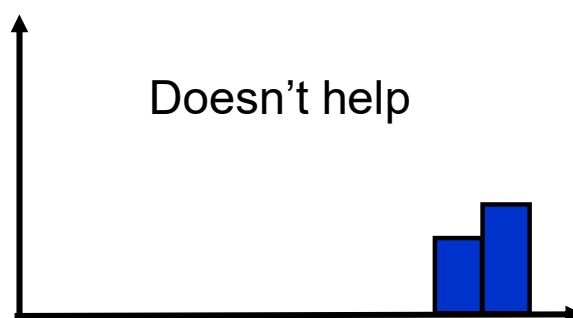
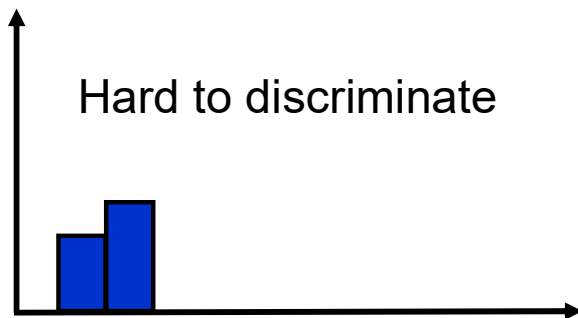
Looking for pixels within a certain hue range.

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Histogram Equalization

- Visual contrast between objects depends on the their gray-level separation.
- How can we improve the contrast **after** an image has been acquired?

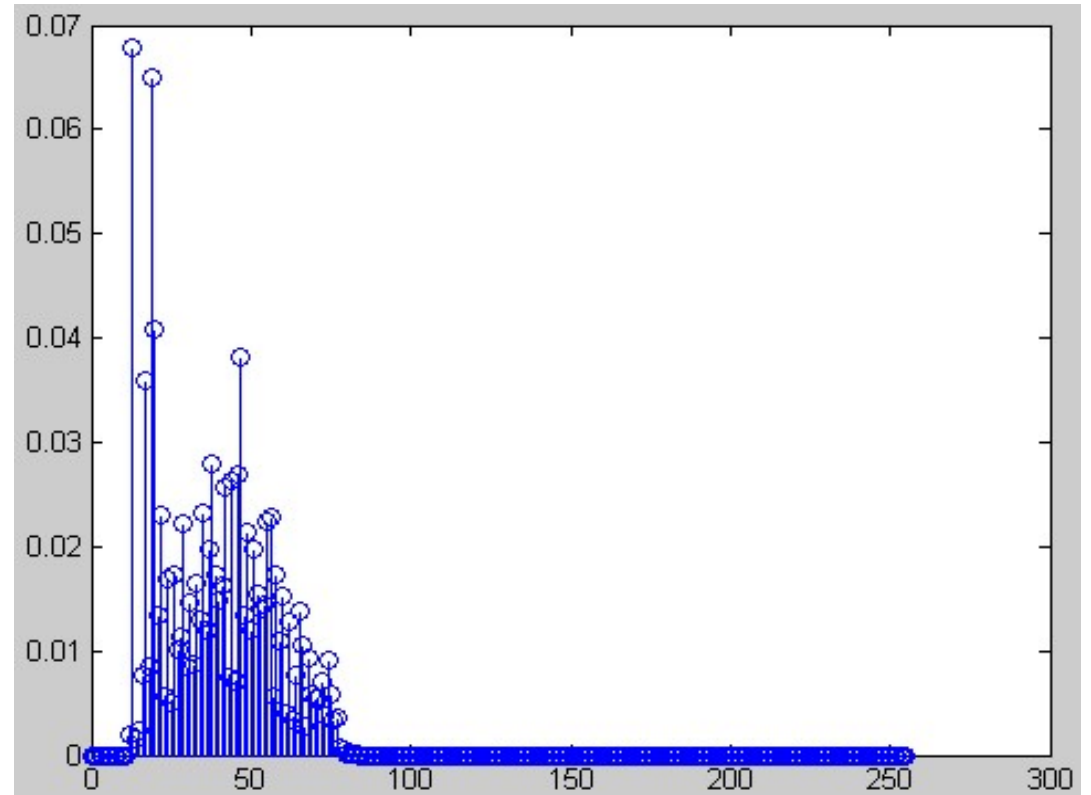
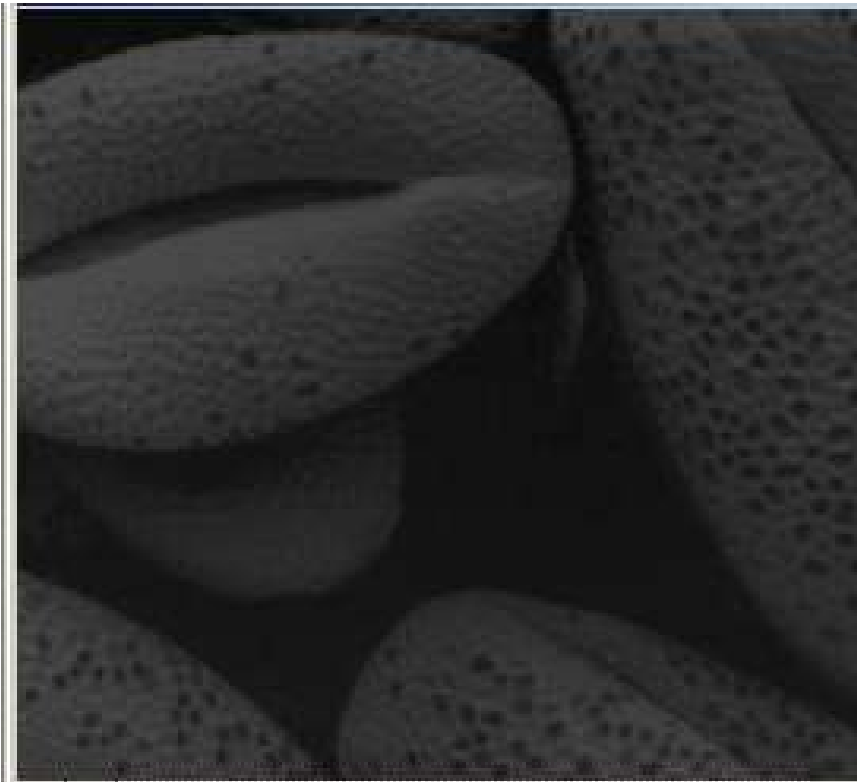


Histogram Equalization

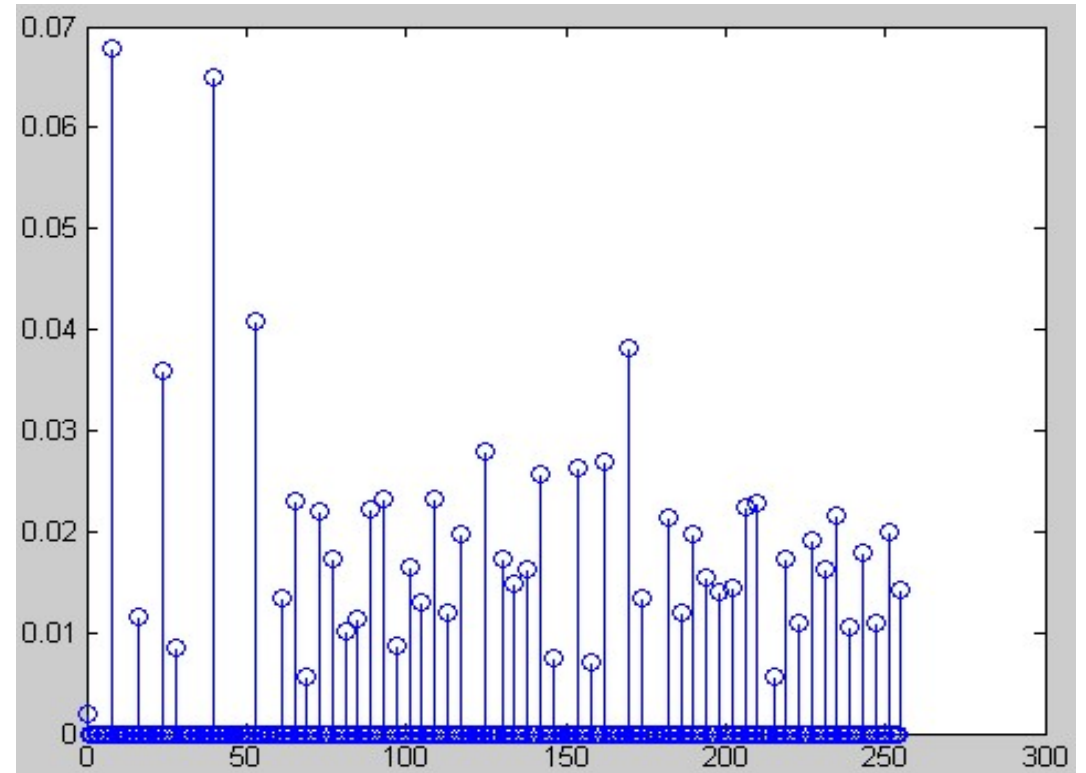
- For a better visual discrimination of an image we would like to re-assign gray-levels so that gray-level resource will be optimally assigned.
- **Our goal:** finding a gray-level transformation $M(v)$ such that:
 - The histogram H_b is as flat as possible.
 - The order of gray-levels is maintained.
 - The histogram bars are not fragmented.

Example

Original image and its histogram

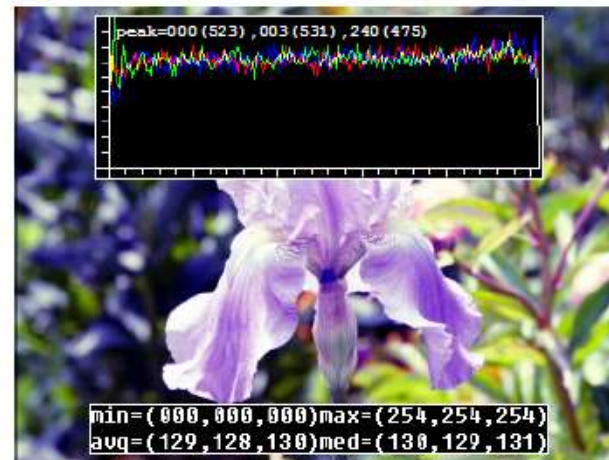
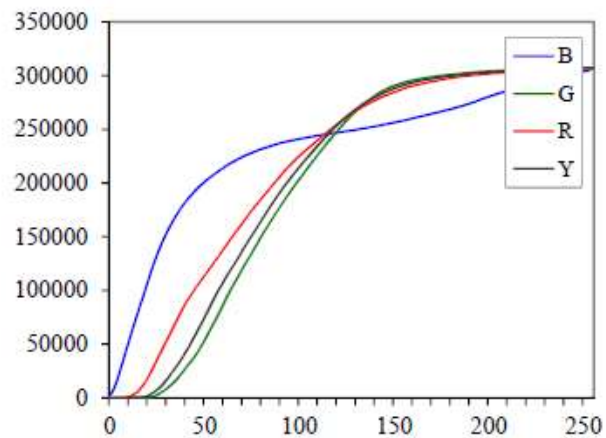
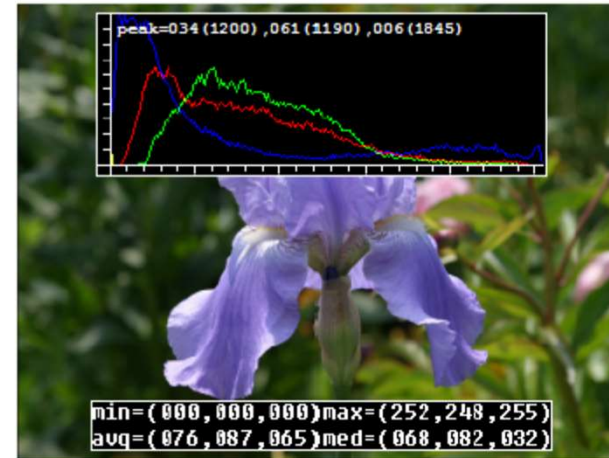
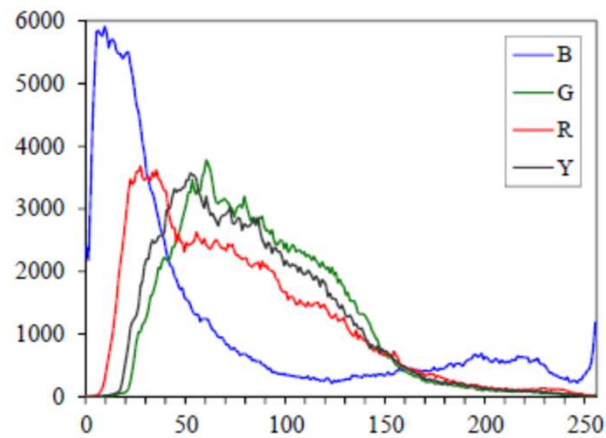


Histogram equalized image and its histogram



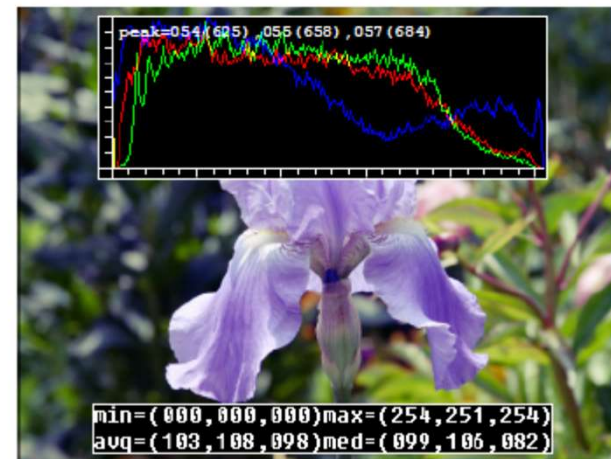
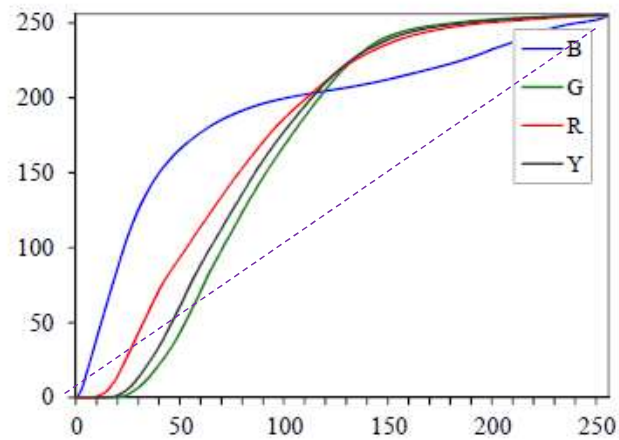
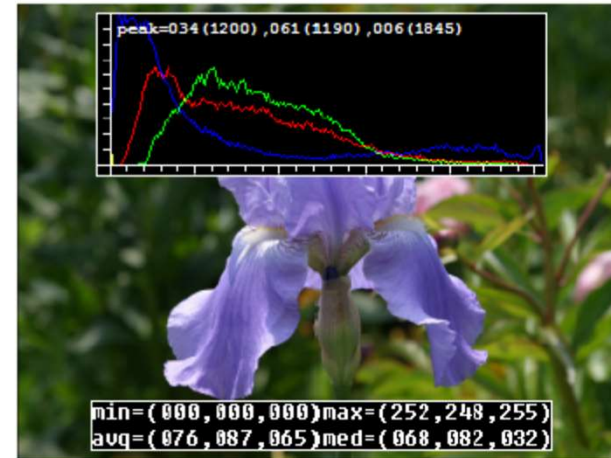
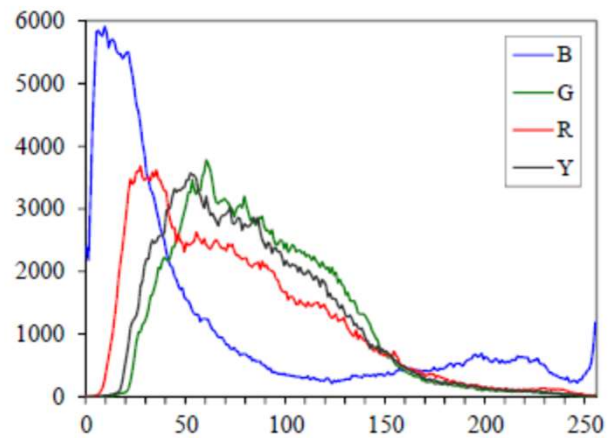
$$c(I) = \frac{1}{N} \sum_{i=0}^I h(i) = c(I-1) + \frac{1}{N} h(I),$$

Cumulative distribution function



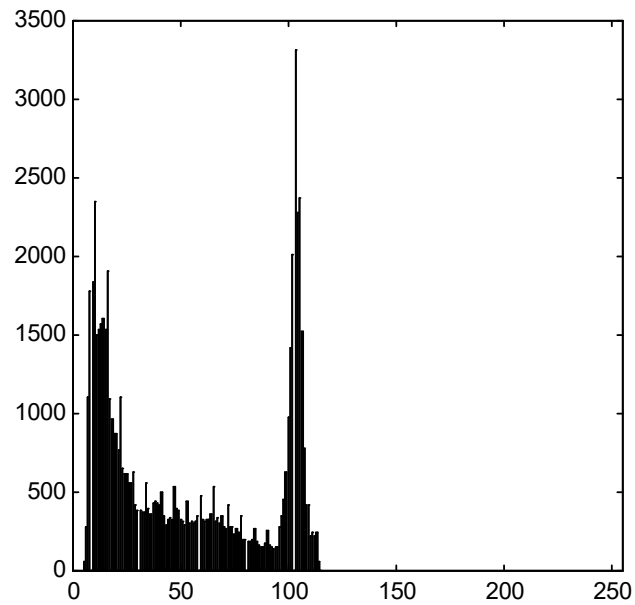
$$c(I) = \frac{1}{N} \sum_{i=0}^I h(i) = c(I-1) + \frac{1}{N} h(I),$$

Cumulative distribution function

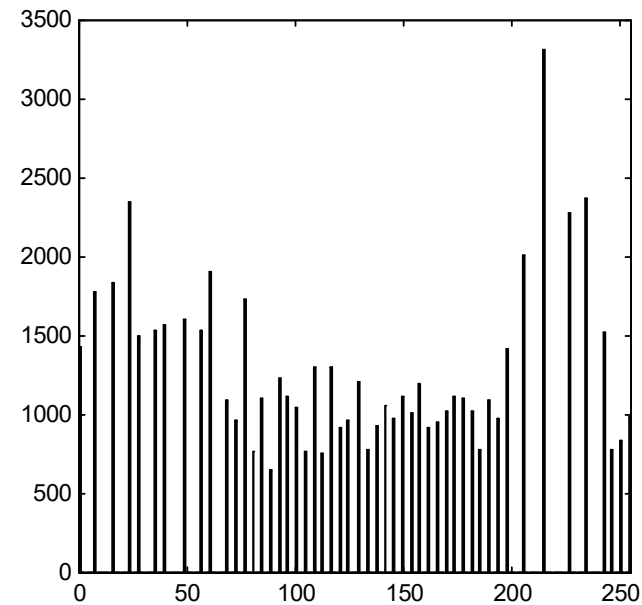


$$\text{Partial } c(I) = \frac{1}{N} \sum_{i=0}^I h(i) = c(I-1) + \frac{1}{N} h(I),$$

Hist. Eq.: Example 1

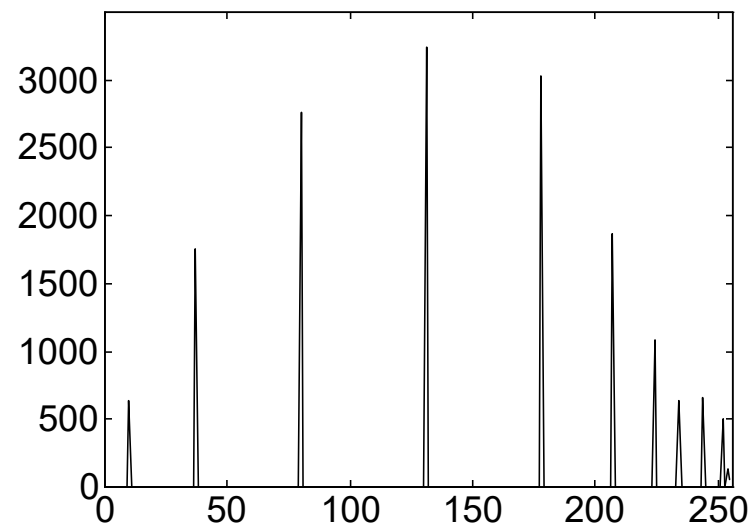
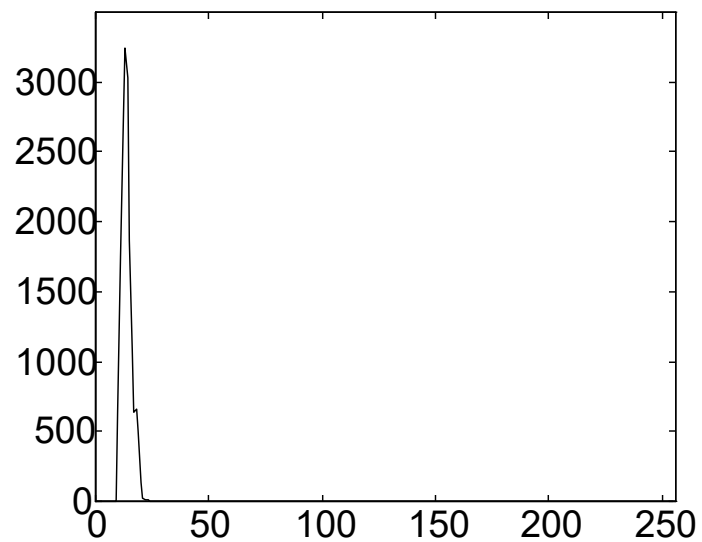


Original



Equalized

Hist. Eq.: Example 2

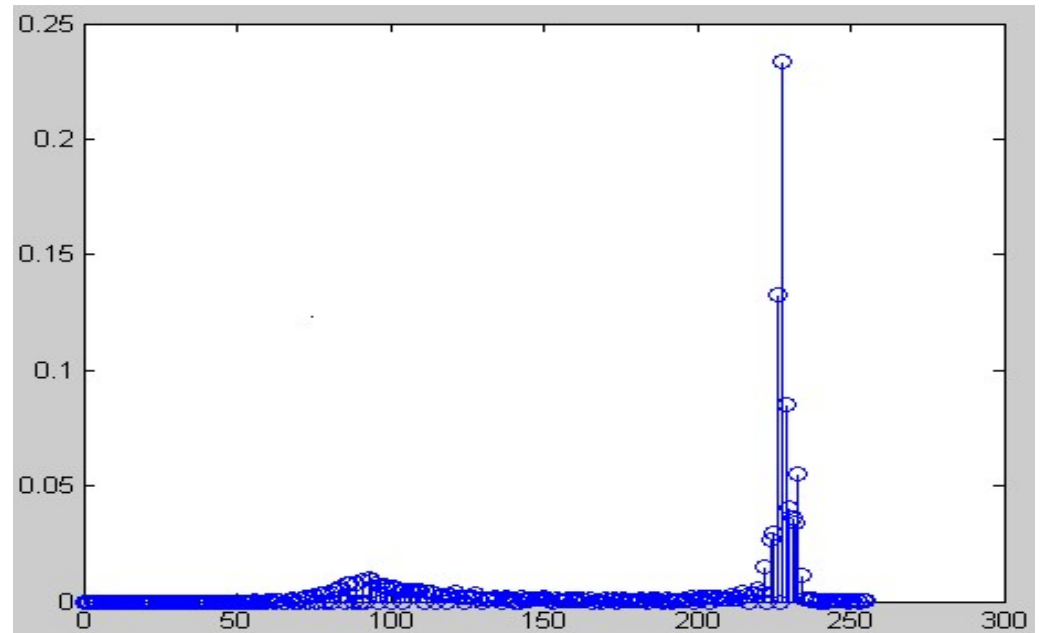


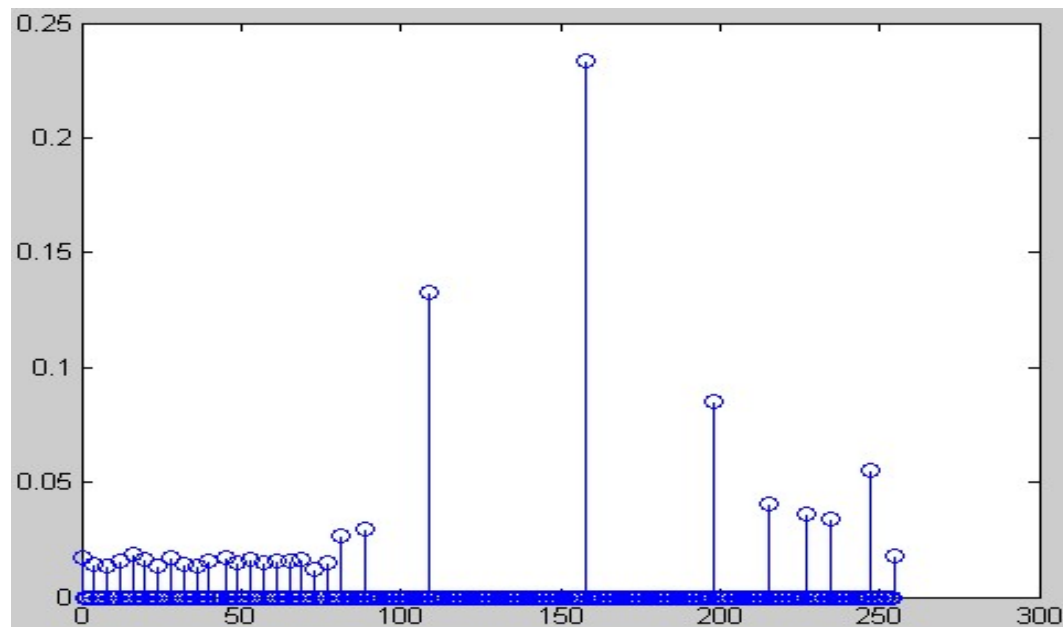
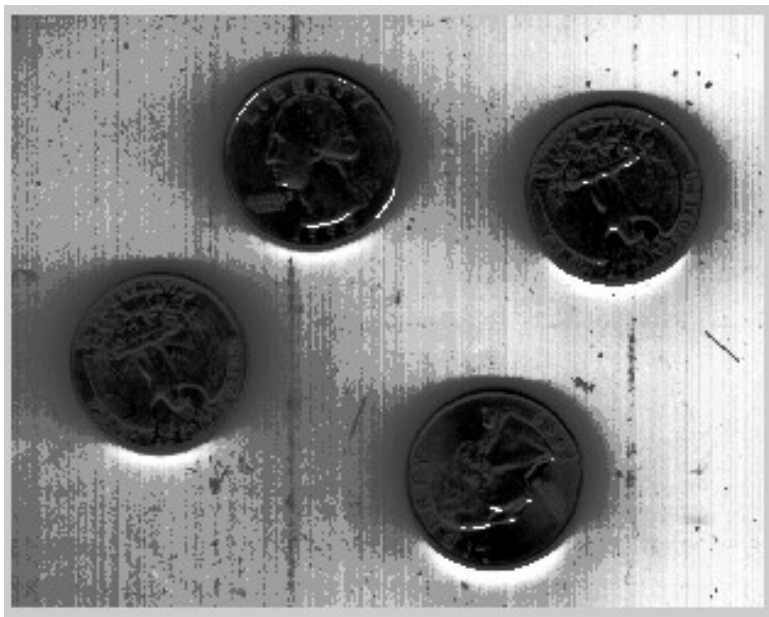
Original

Equalized

- Comments:

Histogram equalization may not always produce desirable results, particularly if the given histogram is very narrow. It can produce false edges and regions. It can also increase image “graininess” and “patchiness.”





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Total Relighting: Learning to Relight Portraits for Background Replacement



Rohit Pandey*, Sergio Orts Escolano*, Chloe LeGendre*, Christian Haene,
Sofien Bouaziz, Christoph Rhemann, Paul Debevec, and Sean Fanello



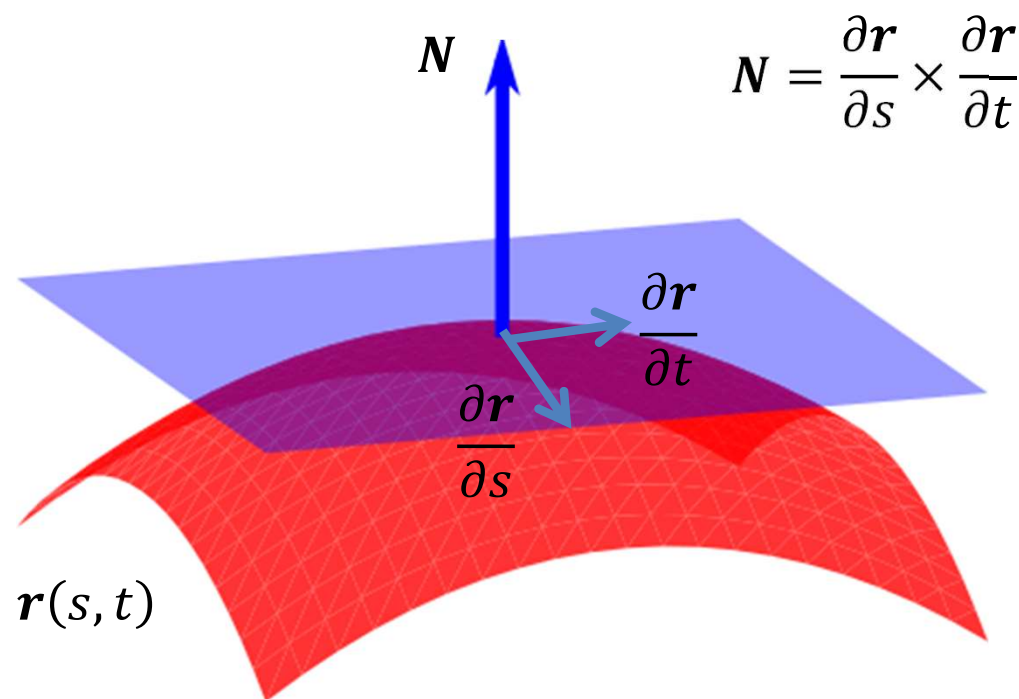
Surface Normal

Parametered 3D surface and its normal

If a (possibly non-flat) surface S in 3D-space $\mathbb{R}^3$ is [parameterized](#) by a system of [curvilinear coordinates](#)

$\mathbf{r}(s, t) = (x(s, t), y(s, t), z(s, t))$ with $(s, t) \in \mathbb{R}$,

then a normal to S is a normal to a tangent plane, given by the cross product of the [partial derivatives](#)

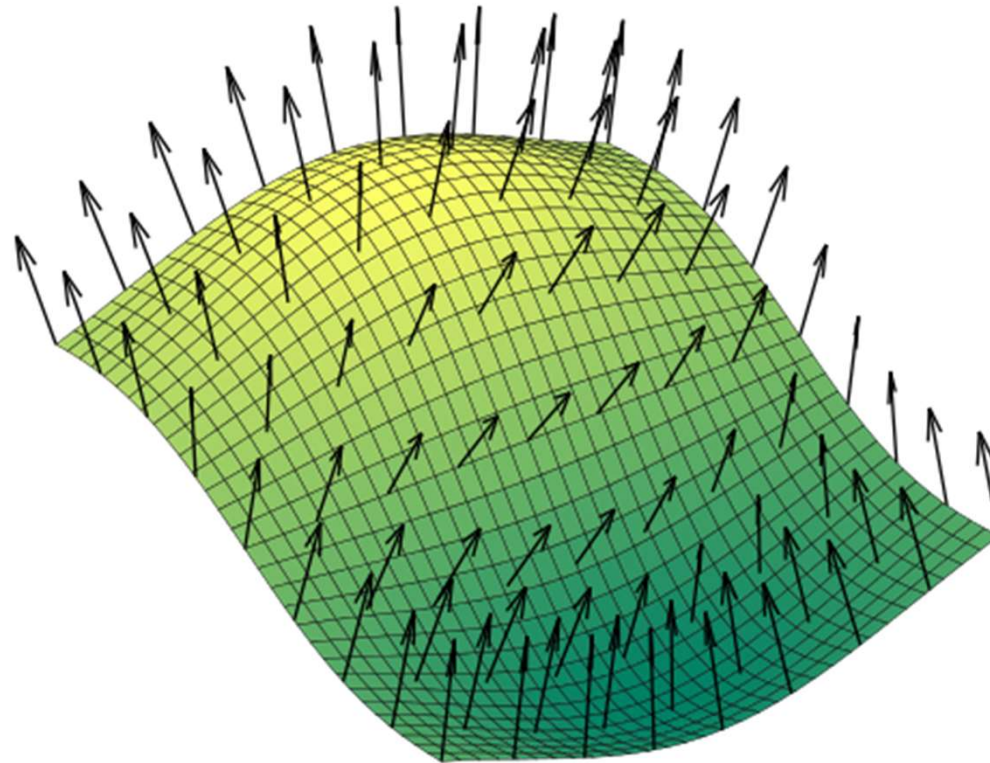


Surface Normal

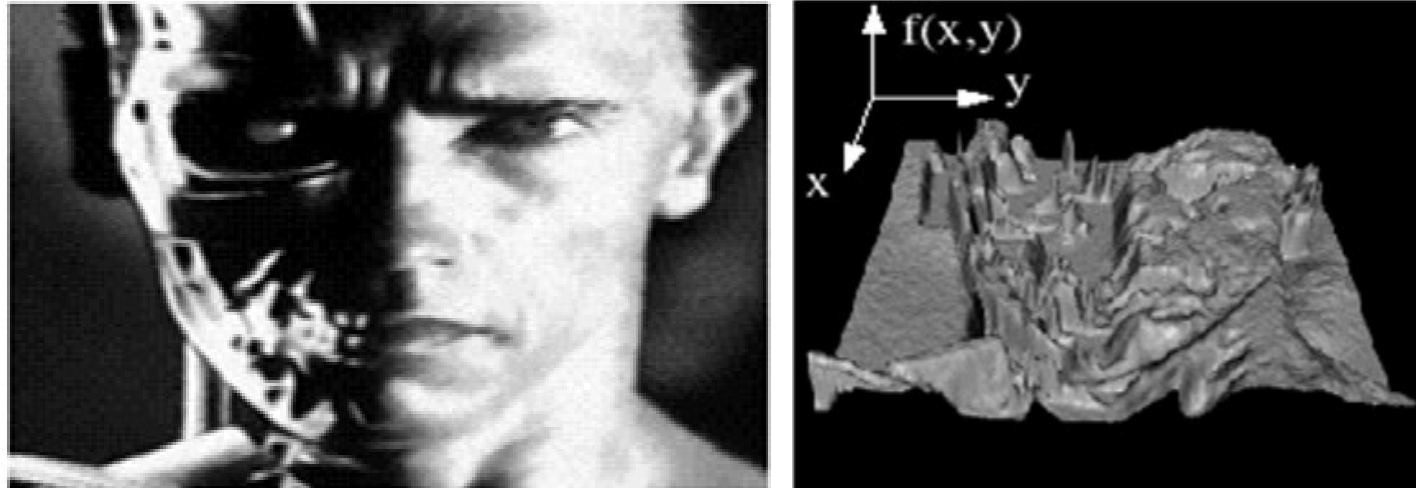
- If a surface $\mathcal{S}$ is given implicitly as the set of points (x, y, z) satisfying $F(x, y, z) = 0$, then a normal at a point (x, y, z) on the surface is given by the gradient:

$$\mathbf{N} = \nabla F(x, y, z)$$

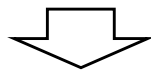
Vector Field of Surface Normal



Surface in Image Coordinates

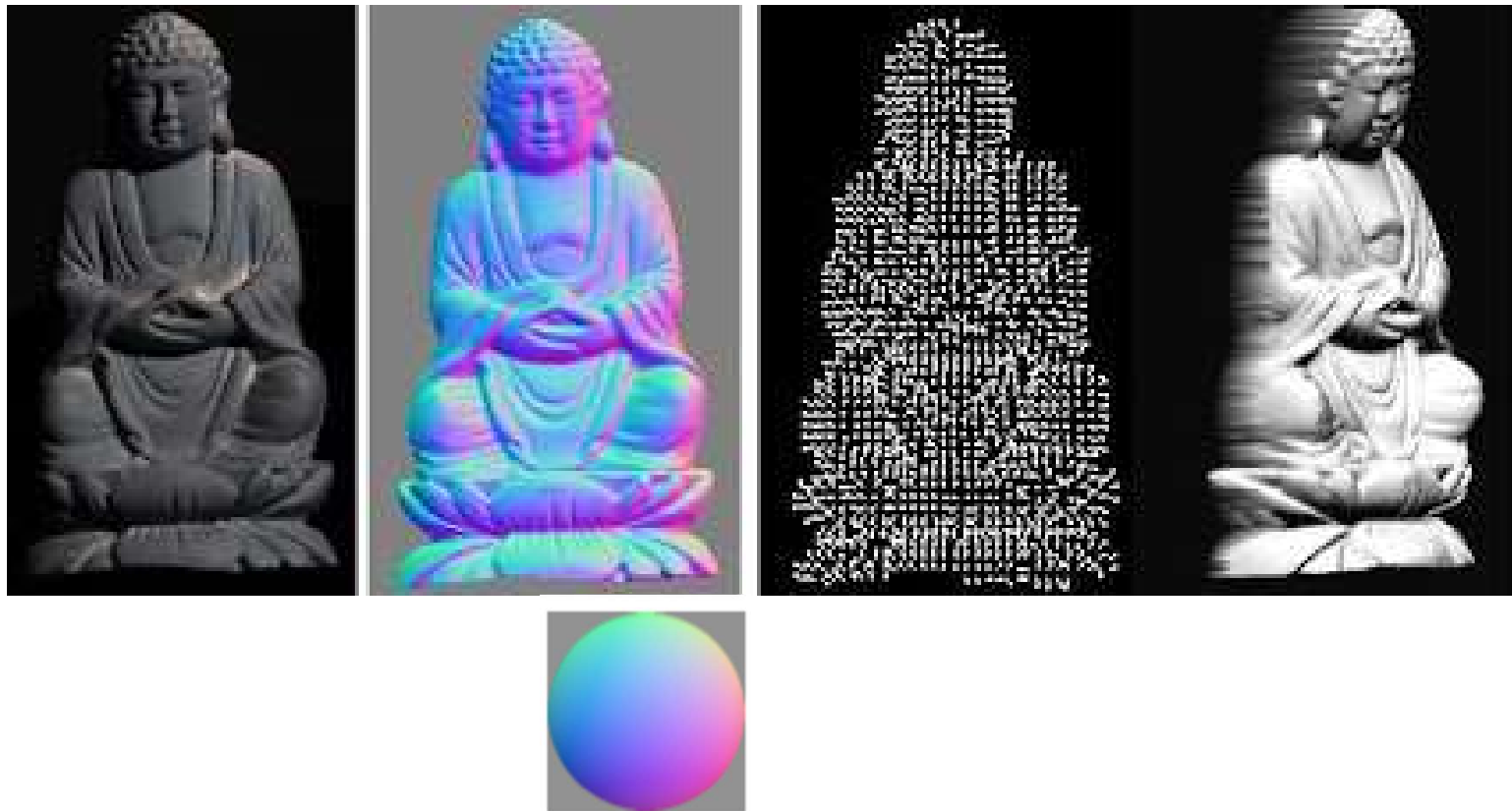


$$\mathbf{r}(s, t) = (x(s, t), y(s, t), z(s, t)) \text{ with } (s, t) \in \mathbb{R}$$

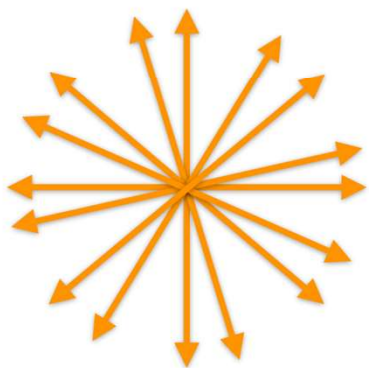


$$\mathbf{r}(x, y) = (x, y, f(x, y)) \text{ with } (x, y) \in \mathbb{Z}$$

Vector Field of Surface Normal of An Image



Light sources

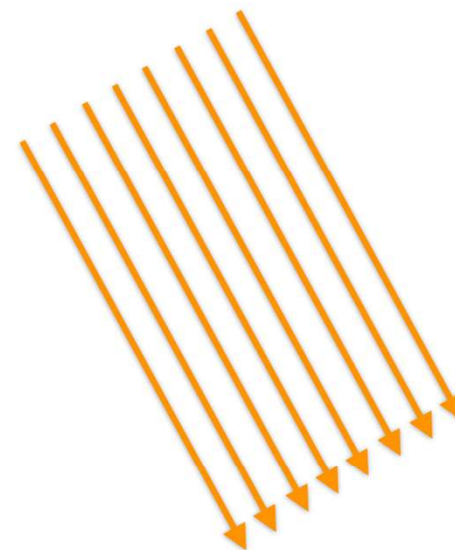


Point Light



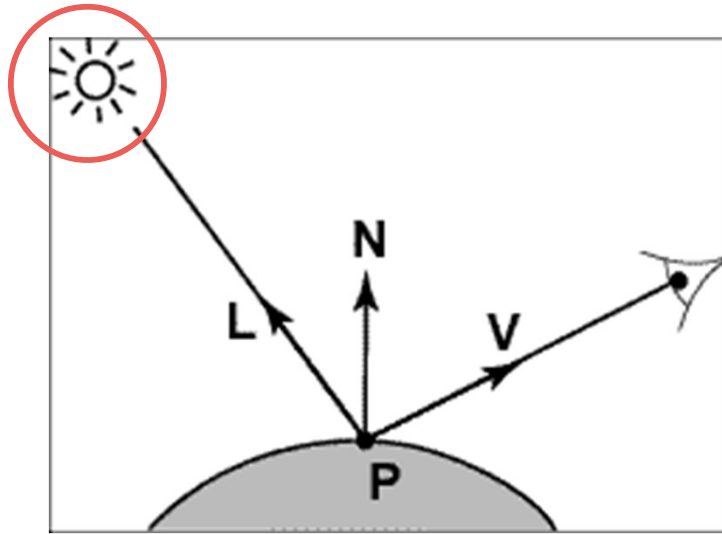
Spot Light

$$L_i(\mathbf{v}_i; \lambda)$$



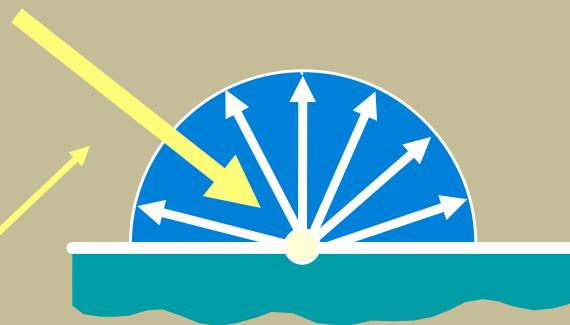
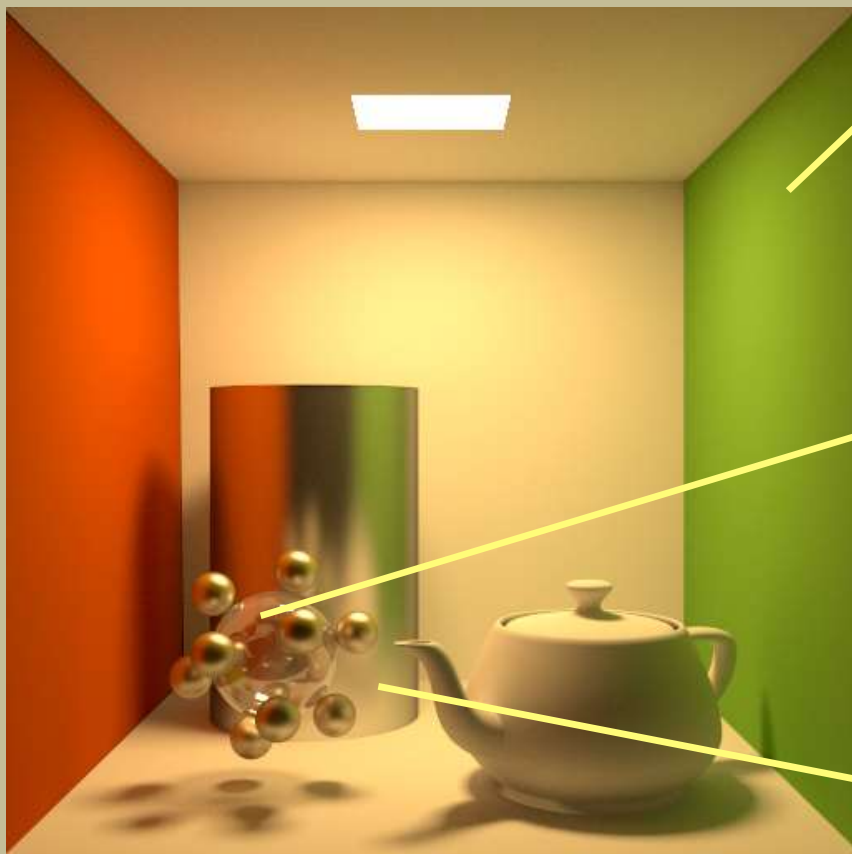
Directional Light

Directional Lighting and Reflectance

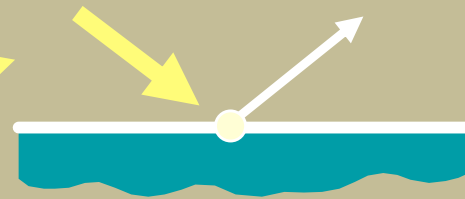


- Key property: all rays are parallel
- Equivalent to an infinitely distant point source

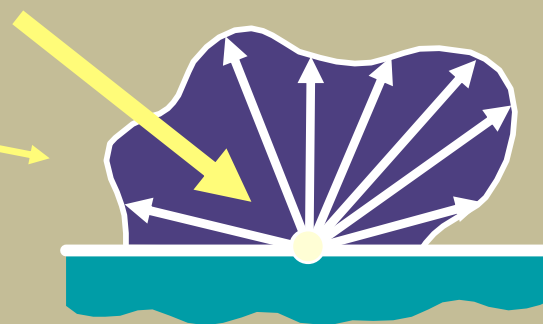
Materials - Three Forms



Ideal diffuse
(Lambertian)

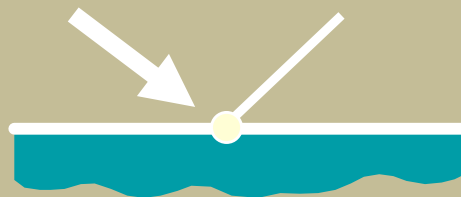
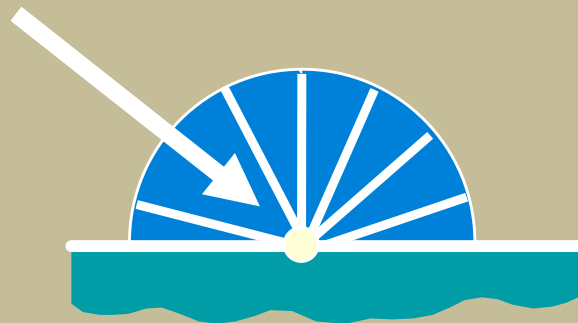


Ideal
specular



Directional
diffuse

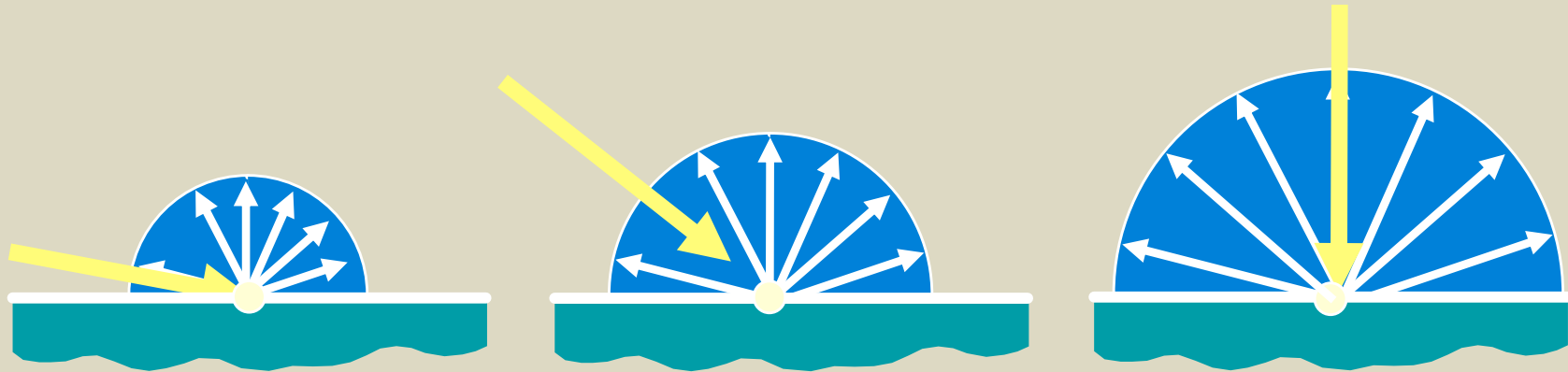
Reflectance—Three Forms



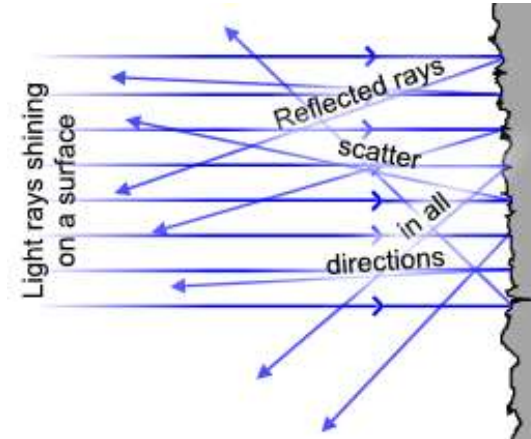
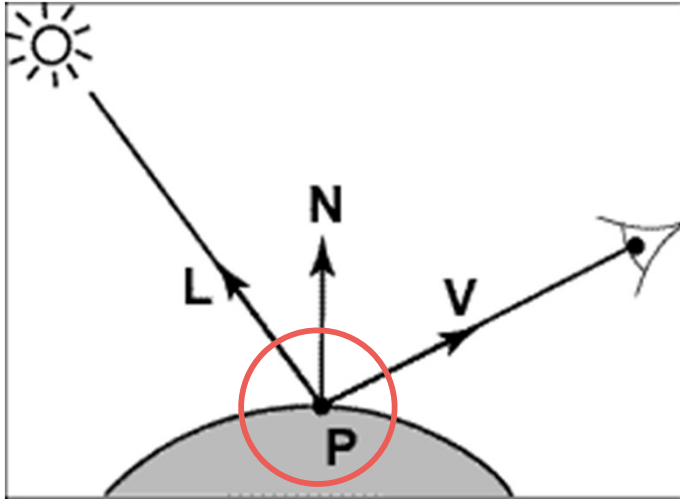
Outline

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 - Histogram Equalization
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Ideal Diffuse: Outgoing Light



Lambertian Reflectance



$$I \propto N \cdot L$$

Image
intensity

$\propto$

Surface
normal

$\cdot$

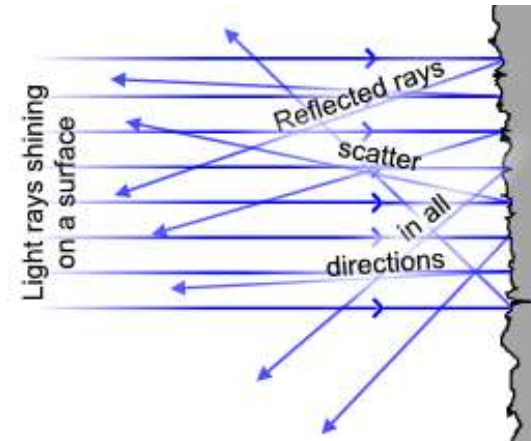
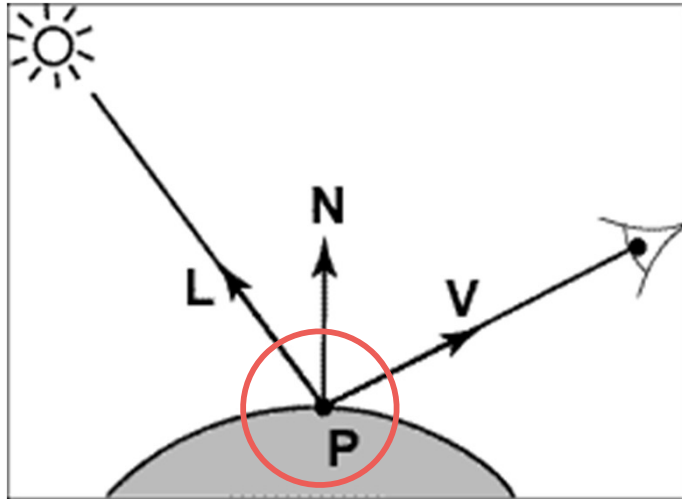
Light
direction

Image
intensity

$\propto$

$\cos(\text{angle between } N \text{ and } L)$

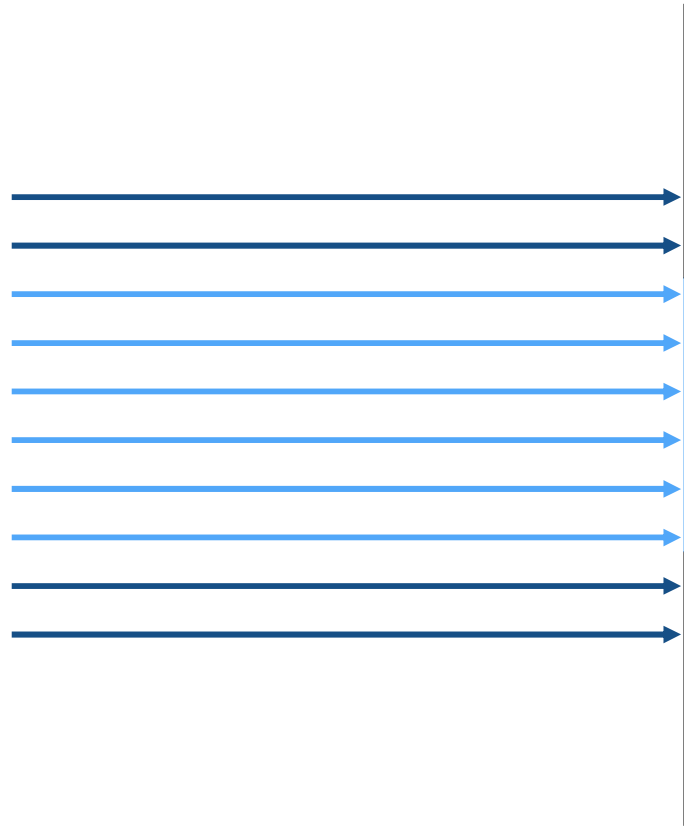
Lambertian Reflectance



1. Reflected energy is proportional to cosine of angle between L and N (**incoming**)
2. Measured intensity is viewpoint-independent (**outgoing**)

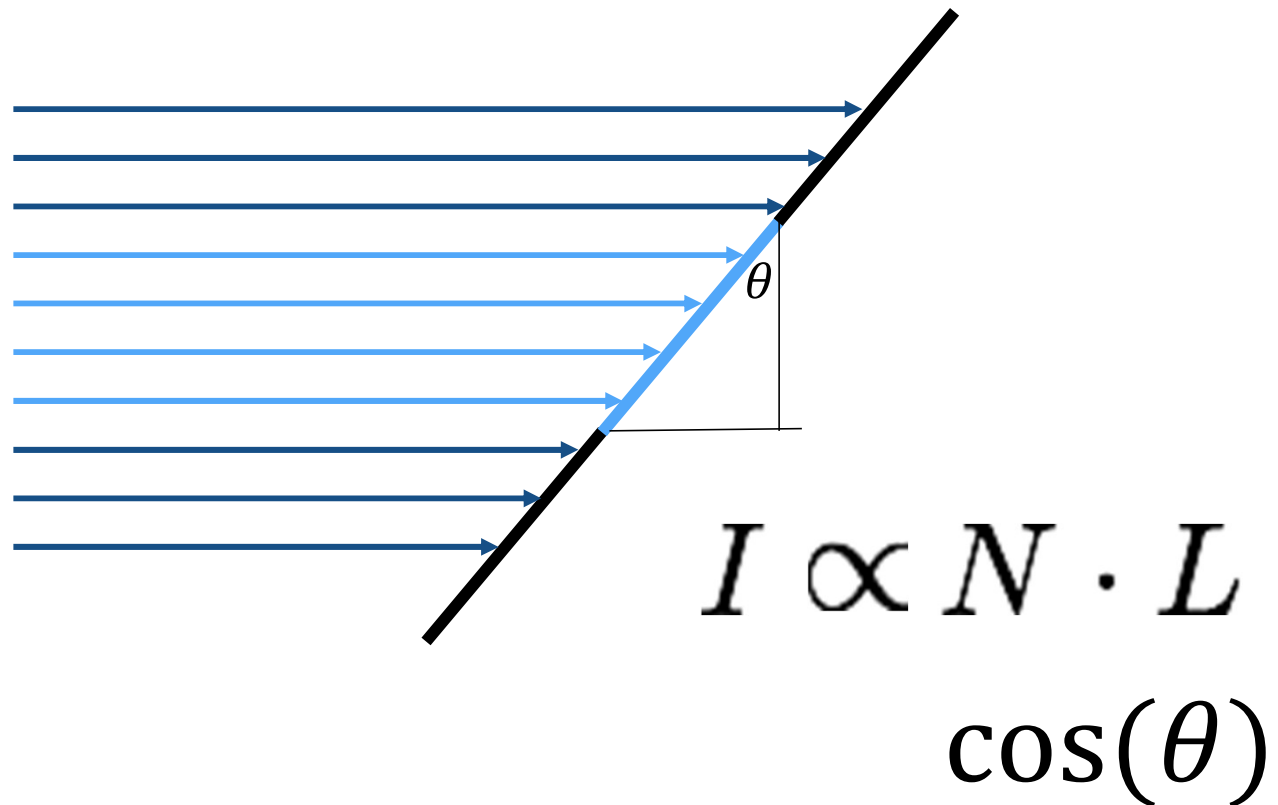
Lambertian Reflectance: Incoming

1. Reflected energy is proportional to cosine of angle between L and N



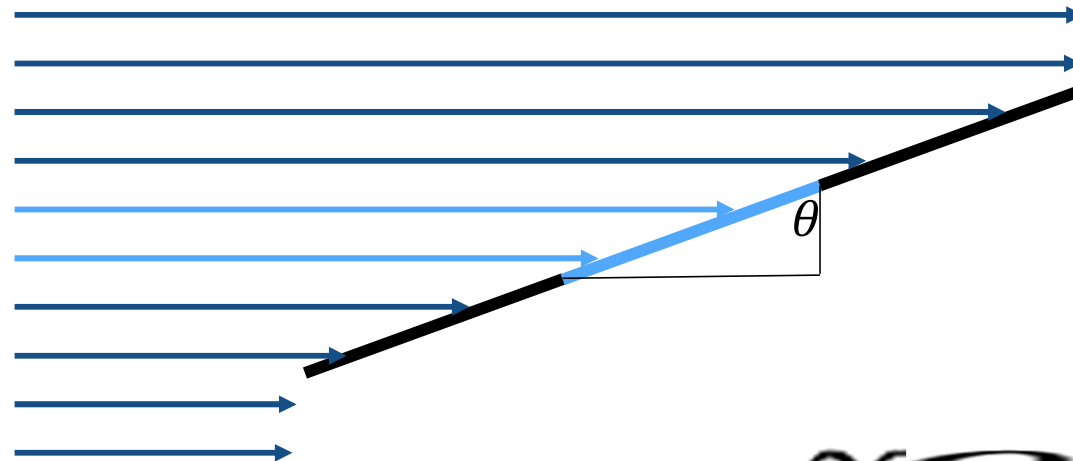
Lambertian Reflectance: Incoming

1. Reflected energy is proportional to cosine of angle between L and N



Lambertian Reflectance: Incoming

1. Reflected energy is proportional to cosine of angle between L and N



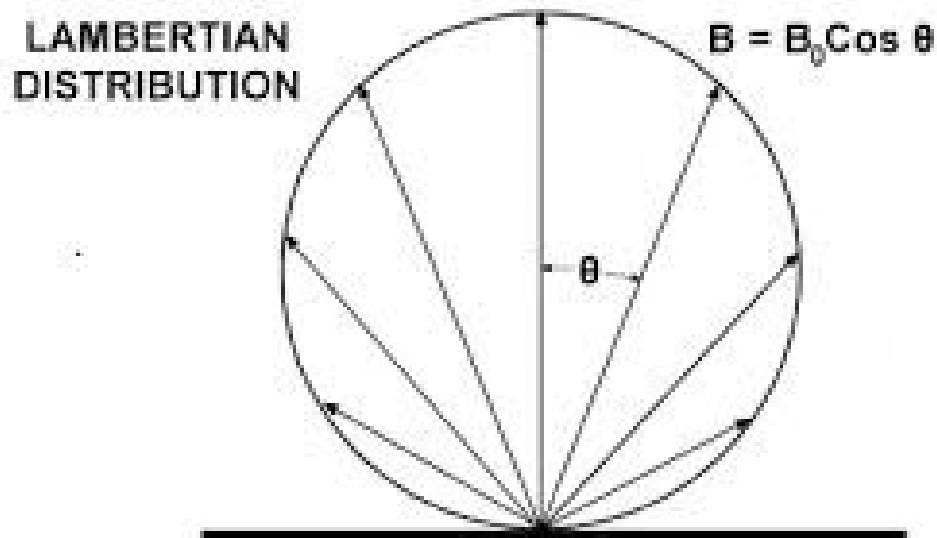
$\propto$

$\cos(\theta)$

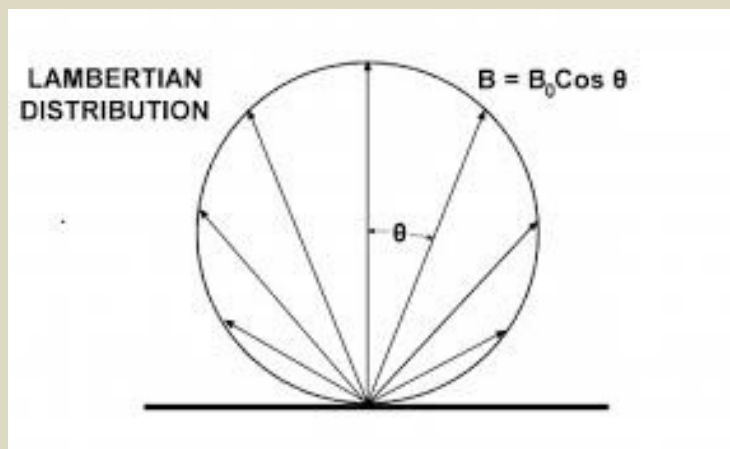
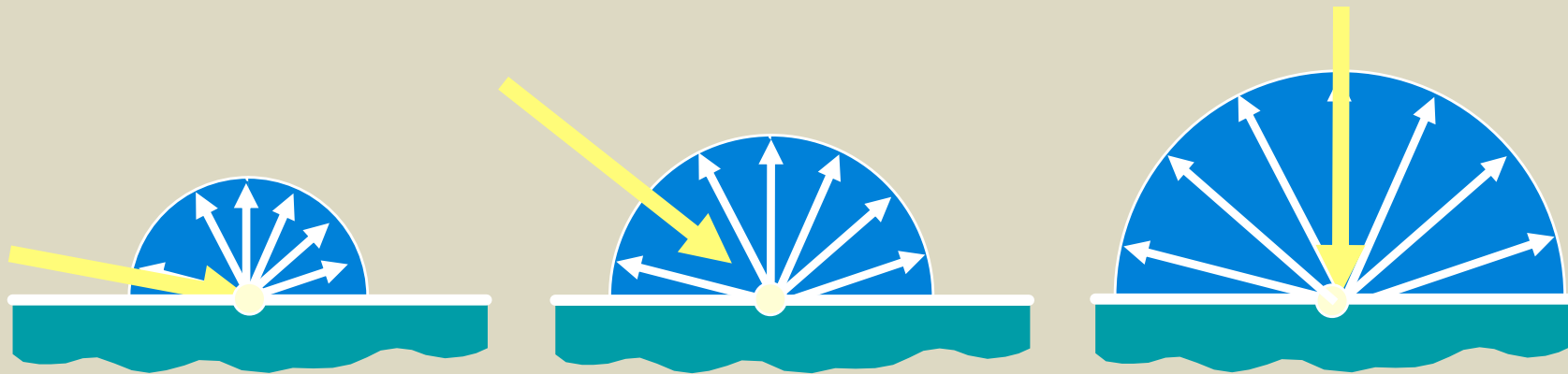
Lambertian Reflectance: Outgoing

1. Radiance (what we see) is viewpoint-independent

$$I \propto N \cdot L$$

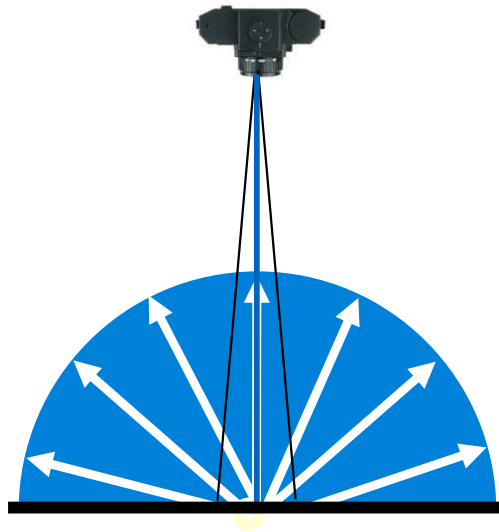


Ideal Diffuse: Outgoing Light



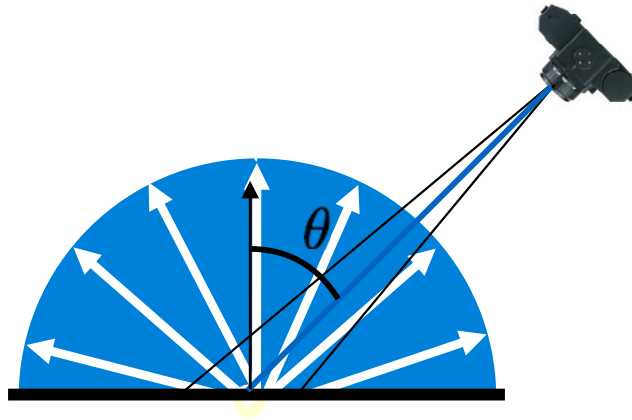
Lambertian Reflectance: Outgoing

1. Radiance (what the eye sees) is viewpoint-independent



Lambertian Reflectance: Outgoing

1. Measured intensity is viewpoint-independent



Lambertian Reflectance: Outgoing

1. Measured intensity is viewpoint-independent

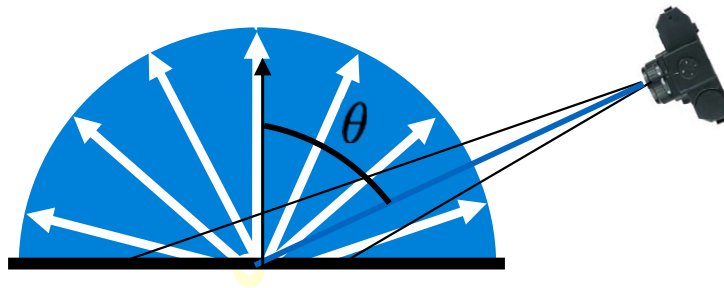
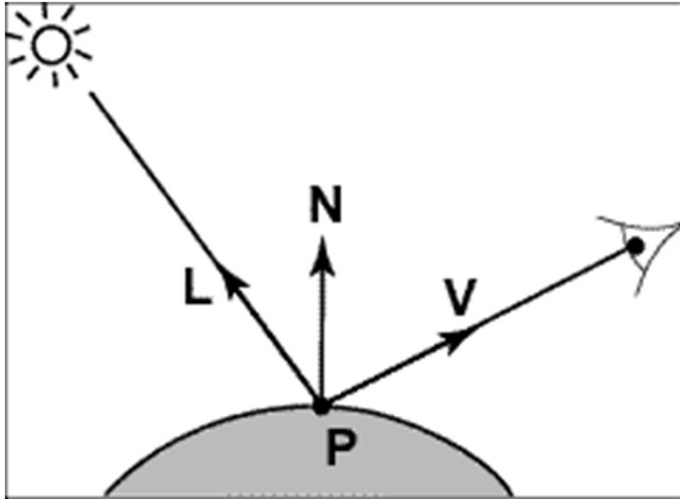


Image Formation Model: Final

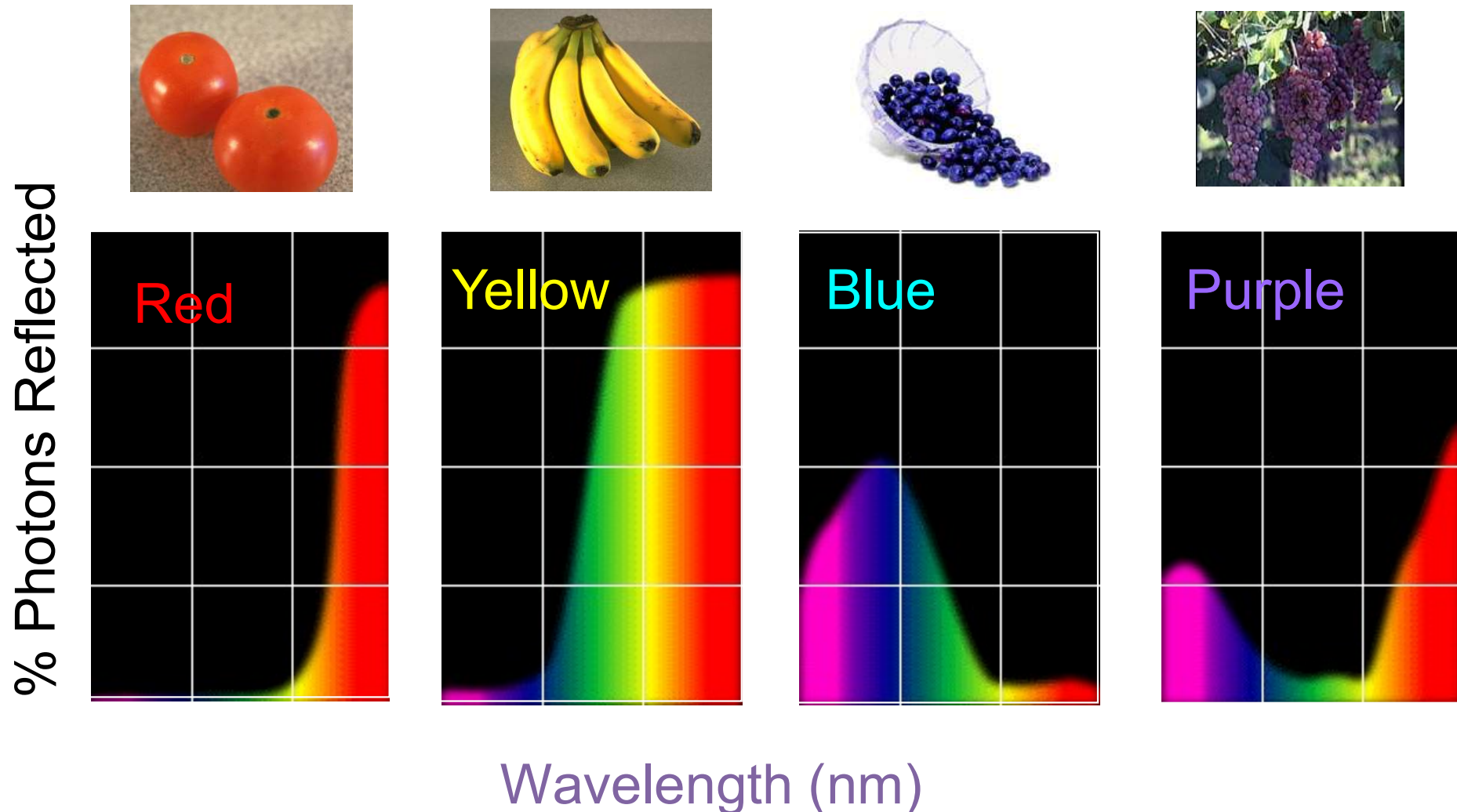


$$I = k_d \mathbf{N} \cdot \mathbf{L}$$

1. Diffuse albedo: what fraction of incoming light is reflected?
 - Introduce scale factor k_d
2. Light intensity: how much light is arriving?
 - Compensate with camera exposure (global scale factor)
3. Camera response function
 - Assume pixel value is linearly proportional to incoming energy (perform radiometric calibration if not)

Reminder from Lecture 2

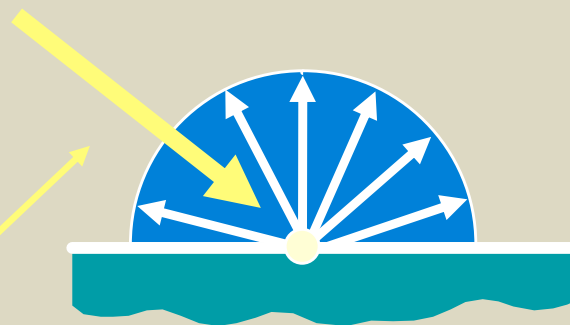
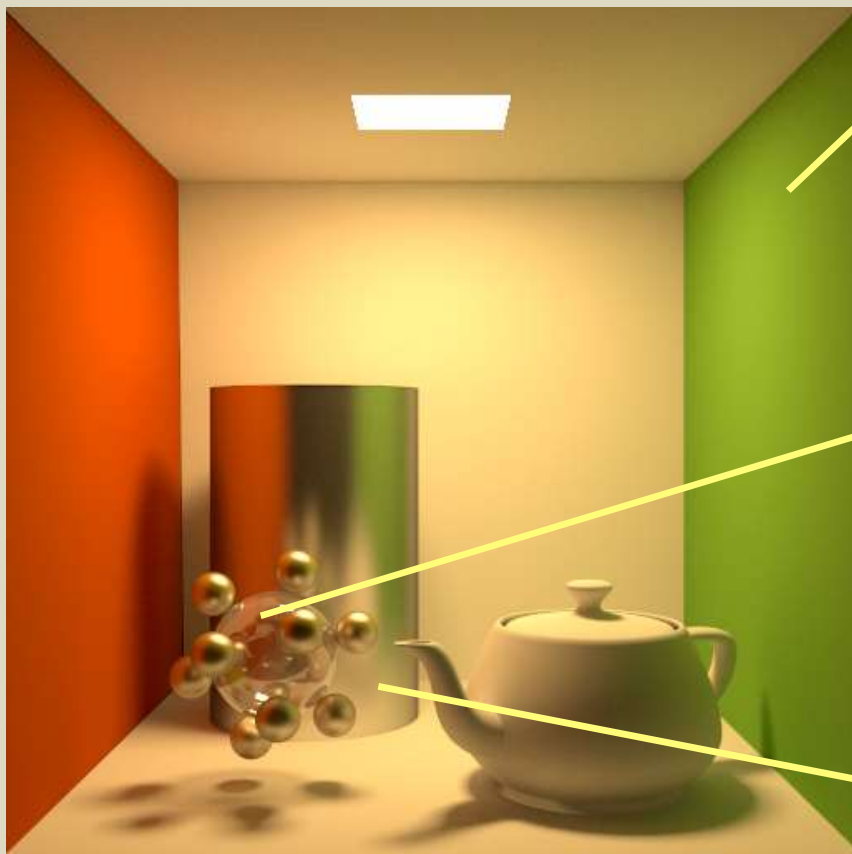
Albedo is wavelength depended.



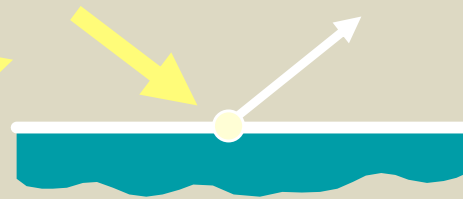
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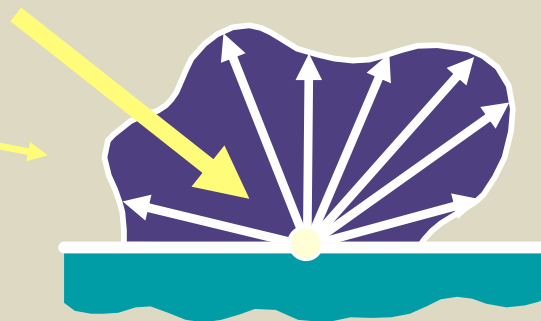
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Ideal diffuse
(Lambertian)

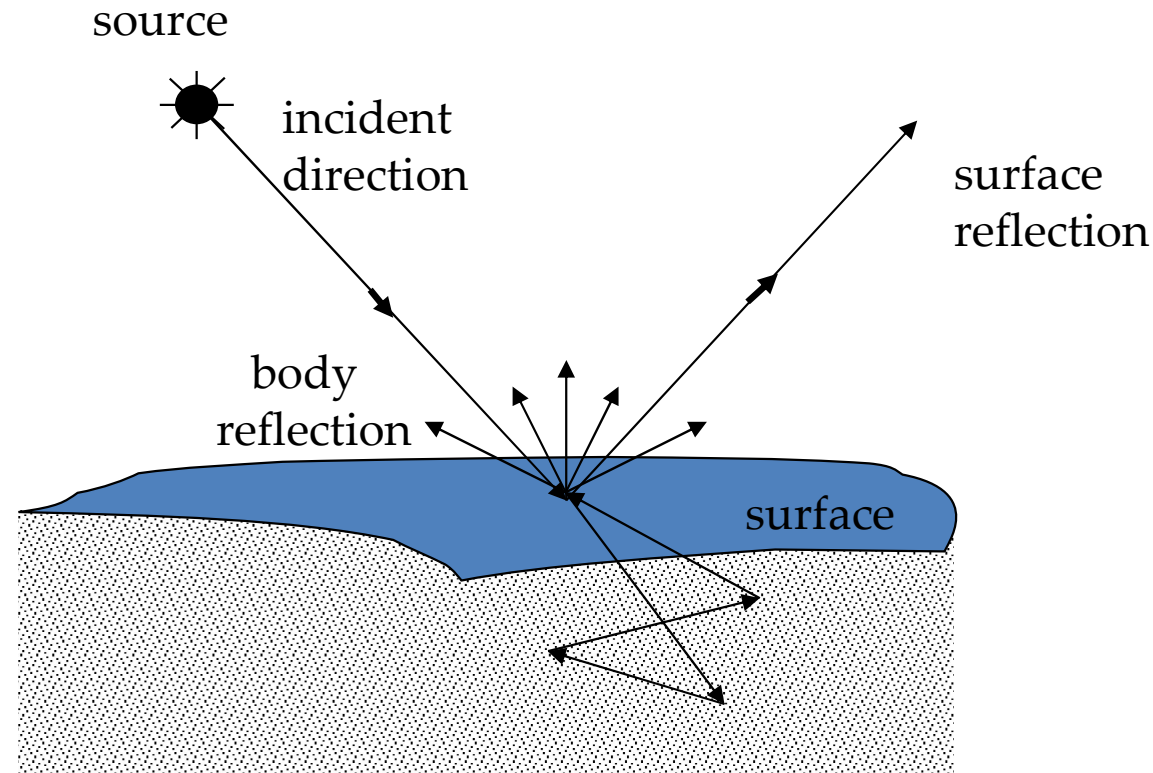


Ideal
specular



Directional
diffuse

Mechanisms of Surface Reflection



Body Reflection:

Diffuse Reflection
Matte Appearance
Non-Homogeneous Medium
Clay, paper, etc

Surface Reflection:

Specular Reflection
Glossy Appearance
Highlights
Dominant for Metals

$$\text{Image Intensity} = \text{Body Reflection} + \text{Surface Reflection}$$

Mechanisms of Surface Reflection

Body Reflection:

Diffuse Reflection
Matte Appearance
Non-Homogeneous Medium
Clay, paper, etc



Surface Reflection:

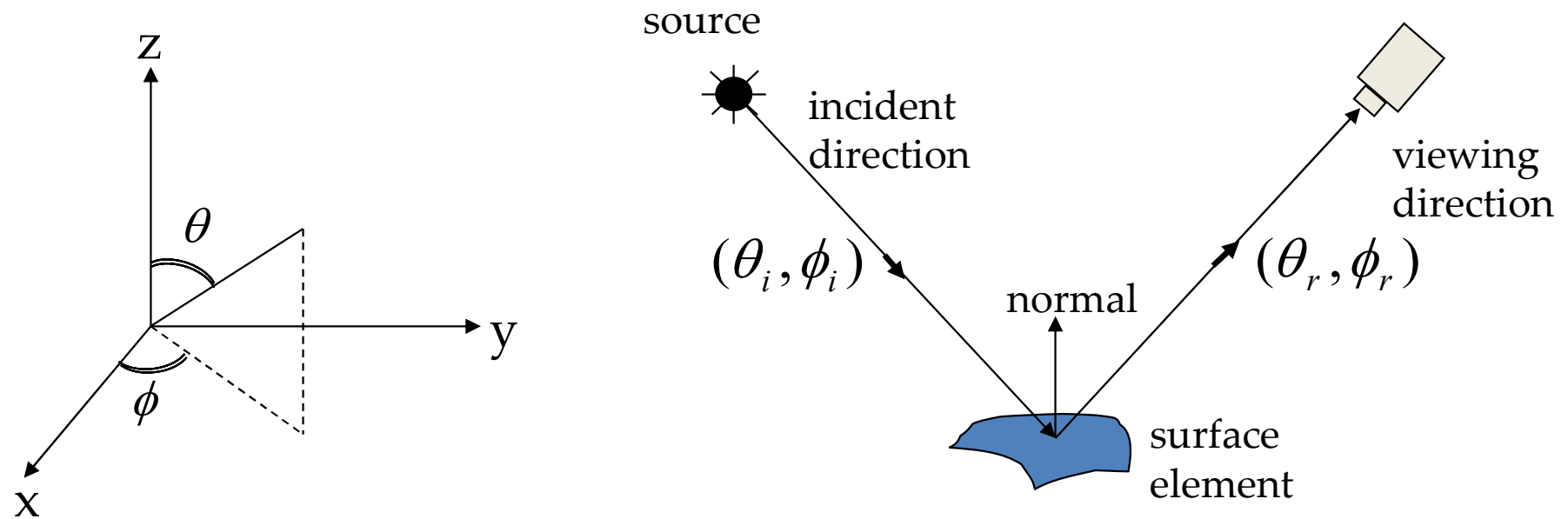
Specular Reflection
Glossy Appearance
Highlights
Dominant for Metals



Many materials exhibit both Reflections:



BRDF: Bidirectional Reflectance Distribution Function



$E^{surface}(\theta_i, \phi_i)$ Irradiance at Surface in direction (θ_i, ϕ_i)

$L^{surface}(\theta_r, \phi_r)$ Radiance of Surface in direction (θ_r, ϕ_r)

$$\text{BRDF } f(\theta_i, \phi_i; \theta_r, \phi_r) = \frac{L^{surface}(\theta_r, \phi_r)}{E^{surface}(\theta_i, \phi_i)}$$

BRDF databases

- MERL ([Matusik](#) et al.): 100 isotropic, 4 nonisotropic, dense

Measurement



SIGGRAPH2005

- 20-80 million reflectance measurements per material
- Each tabulated BRDF entails $90 \times 90 \times 180 \times 3 = 4,374,000$ measurement bins



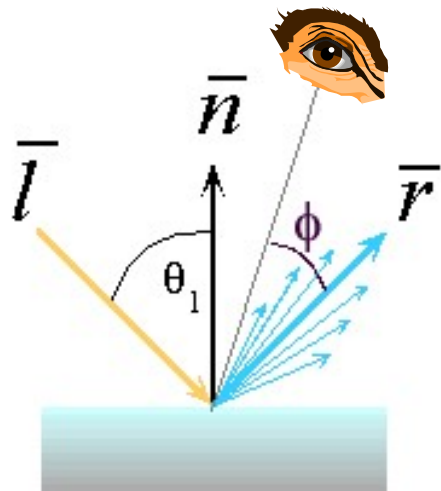
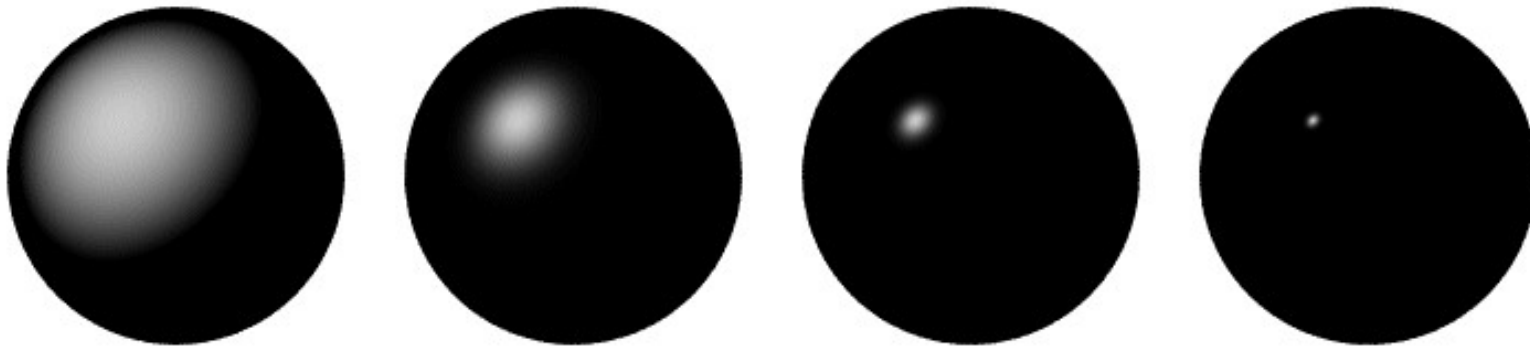
Course 10: Realistic Materials in Computer Graphics

Wojciech Matusik

- [CURET](#) (Columbia-Utrecht): 60 samples, more sparsely sampled, but also bidirectional texture functions (BTF)

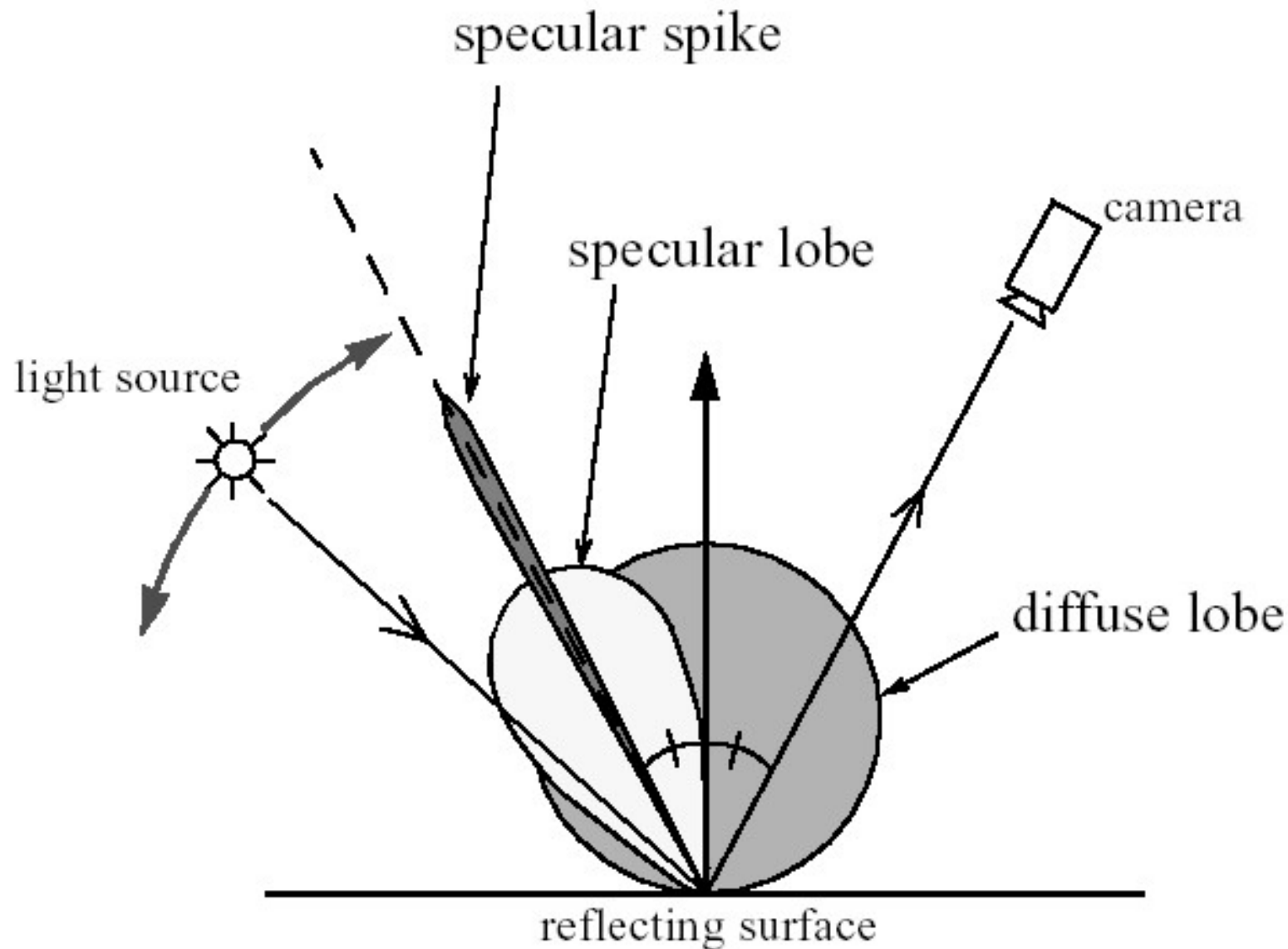
Phong Model

- These spheres illustrate the Phong model as *lighting direction* and n_{shiny} are varied:



$$L = I \rho_s (\cos \phi)^{n_{shiny}}$$

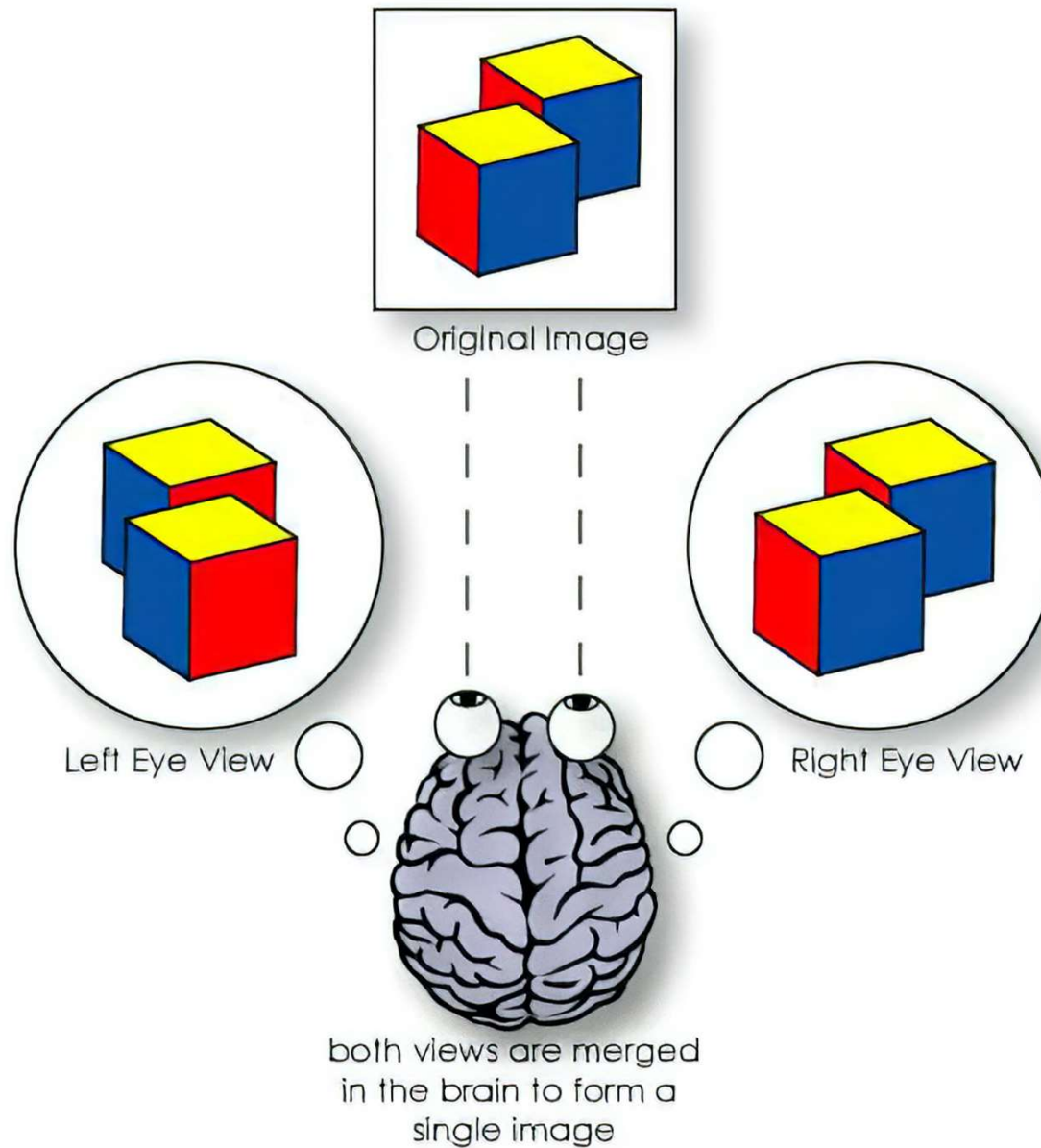
All components of Surface Reflection



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Two view stereo



Two view stereo



Key Idea: use feature motion to understand shape

Human stereo from single image?



Photometric Stereo



Key Idea: use pixel brightness to understand shape

Photometric Stereo



Key Idea: use pixel brightness to understand shape

A Single Image: Shape from Shading



$$I = k_d \mathbf{N} \cdot \mathbf{L}$$

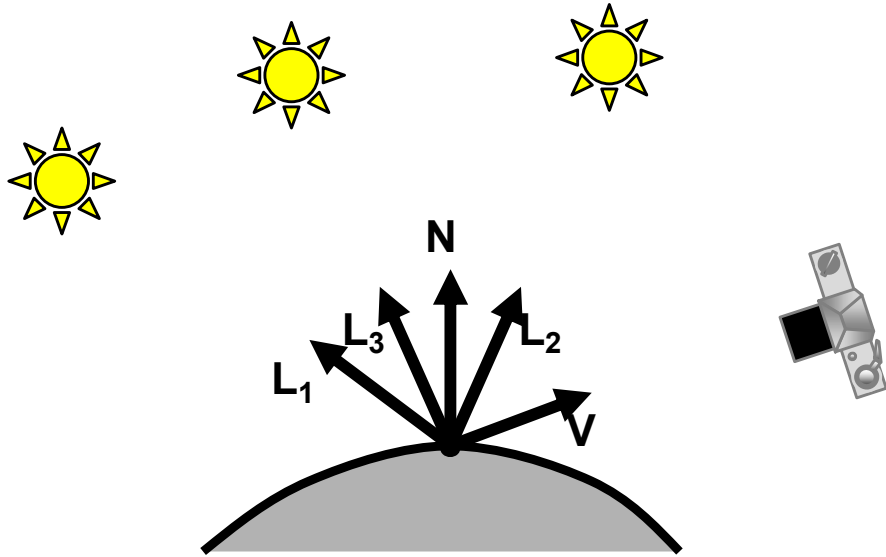
Assume k_d is 1 for now.

What can we measure from one image?

$\cos^{-1}(I)$ is the angle between $\mathbf{N}$ and $\mathbf{L}$

- Add assumptions:
 - A few known normals (e.g., silhouettes)
 - Smoothness of normals

Multiple Images: Photometric Stereo



$$I_1 = k_d \mathbf{N} \cdot \mathbf{L}_1$$

$$I_2 = k_d \mathbf{N} \cdot \mathbf{L}_2$$

$$I_3 = k_d \mathbf{N} \cdot \mathbf{L}_3$$

Write this as a matrix equation:

$$\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix} = k_d \mathbf{N}^T \begin{bmatrix} \mathbf{L}_1 & \mathbf{L}_2 & \mathbf{L}_3 \end{bmatrix}$$

Solving the Equations

$$\underbrace{\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix}}_{\substack{\mathbf{I} \\ 1 \times 3}} = k_d \underbrace{\mathbf{N}^T}_{\substack{\mathbf{G} \\ 1 \times 3}} \underbrace{\begin{bmatrix} \mathbf{L}_1 & \mathbf{L}_2 & \mathbf{L}_3 \end{bmatrix}}_{\substack{\mathbf{L} \\ 3 \times 3}}$$

Solving:

$$\mathbf{G} = \mathbf{I}\mathbf{L}^{-1}$$

Normalized the G, separate it into mode k_d and normal N

$$k_d = \|\mathbf{G}\|$$

$$\mathbf{N} = \frac{1}{k_d} \mathbf{G}$$

Solving the Equations

$$\underbrace{\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix}}_{\substack{\mathbf{I} \\ 1 \times 3}} = k_d \underbrace{\mathbf{N}^T}_{\substack{\mathbf{G} \\ 1 \times 3}} \underbrace{\begin{bmatrix} \mathbf{L}_1 & \mathbf{L}_2 & \mathbf{L}_3 \end{bmatrix}}_{\substack{\mathcal{L} \\ 3 \times 3}}$$

$$\mathbf{G} = \mathbf{I}\mathbf{L}^{-1}$$

- When is $\mathbf{L}$ nonsingular (invertible)?
 - ≥ 3 light directions are linearly independent, or:
 - All light direction vectors cannot lie in a plane.
- What if we have more than one pixel?
 - Stack them all into one big system.

More than Three Lights

$$\begin{bmatrix} I_1 & \dots & I_n \end{bmatrix} = k_d \mathbf{N}^T \begin{bmatrix} \mathbf{L}_1 & \dots & \mathbf{L}_n \end{bmatrix}$$

- Solve using least squares (normal equations):

$$\mathbf{I} = \mathbf{G}\mathbf{L}$$

Find $\mathbf{G}$ s.t. minimizes $\|\mathbf{I} - \mathbf{G}\mathbf{L}\|^2$

$$\mathbf{I}\mathbf{L}^T = \mathbf{G}\mathbf{L}\mathbf{L}^T$$

$$\mathbf{G} = (\mathbf{I}\mathbf{L}^T)(\mathbf{L}\mathbf{L}^T)^{-1}$$

- Given $\mathbf{G}$, solve for $\mathbf{N}$ and k_d as before.

More than one pixel

Previously:

$$\begin{array}{c} 1 \times \# \text{ images} \\ \boxed{\text{I}} \end{array} = \begin{array}{c} 1 \times 3 \\ \boxed{\text{N}} \end{array} * \begin{array}{c} 3 \times \# \text{ images} \\ \boxed{\text{L}} \end{array}$$

More than one pixel

Stack all pixels into one system:

$$\begin{array}{c} p \times \# \text{ images} \\ \boxed{\text{I}} \end{array} = \begin{array}{c} p \times 3 \\ \boxed{\text{N}} \end{array} * \begin{array}{c} 3 \times \# \text{ images} \\ \boxed{\text{L}} \end{array}$$

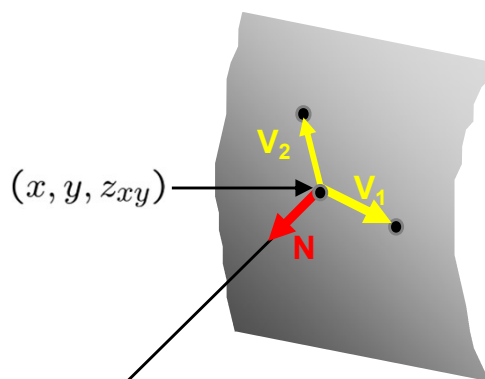
Solve as before.

Depth Map from Normal Map

- We now have a surface normal, but how do we get depth?

Assume a smooth surface

orthographic projection

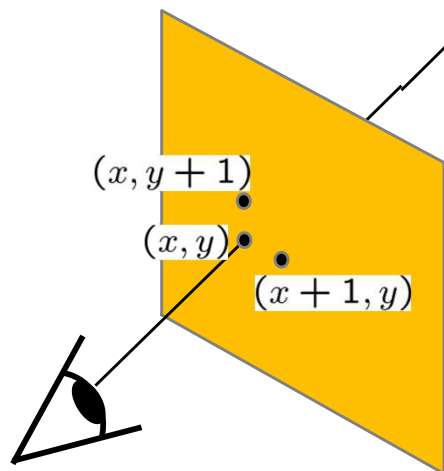


$$\begin{aligned} V_1 &= (x + 1, y, z_{x+1,y}) - (x, y, z_{xy}) \\ &= (1, 0, z_{x+1,y} - z_{xy}) \end{aligned}$$

$$\begin{aligned} 0 &= N \cdot V_1 \\ &= (n_x, n_y, n_z) \cdot (1, 0, z_{x+1,y} - z_{xy}) \\ &= n_x + n_z(z_{x+1,y} - z_{xy}) \end{aligned}$$

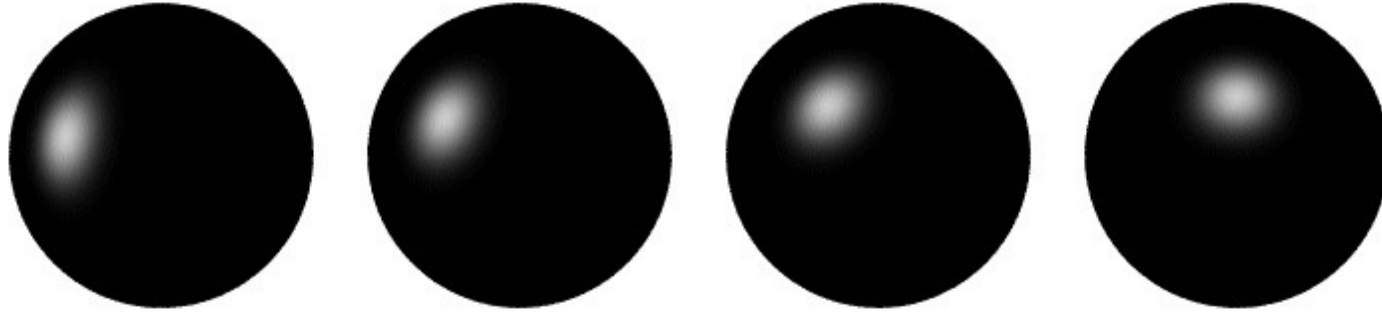
Get a similar equation for V_2

- Each normal gives us two linear constraints on z
- compute z values by solving a matrix equation



Determining Light Directions

- Trick: Place a mirror ball in the scene.



- The location of the highlight is determined by the light source direction.

Mirrored Sphere



HDRI Sky Probe



Real-World HDR Lighting Environments

Funston
Beach



Eucalyptus
Grove



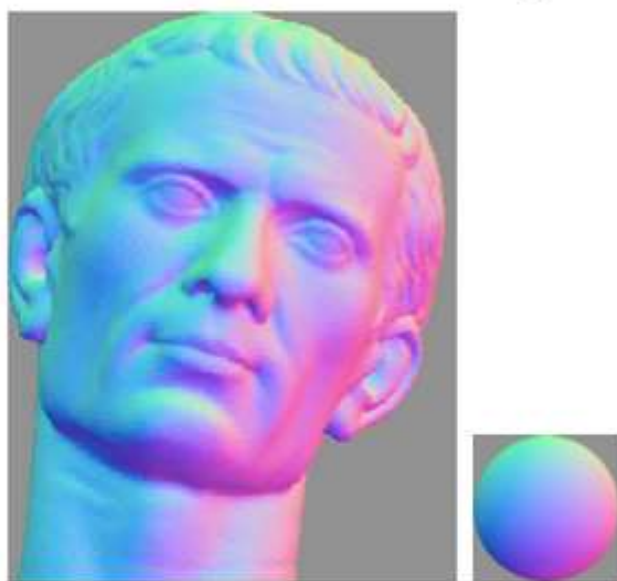
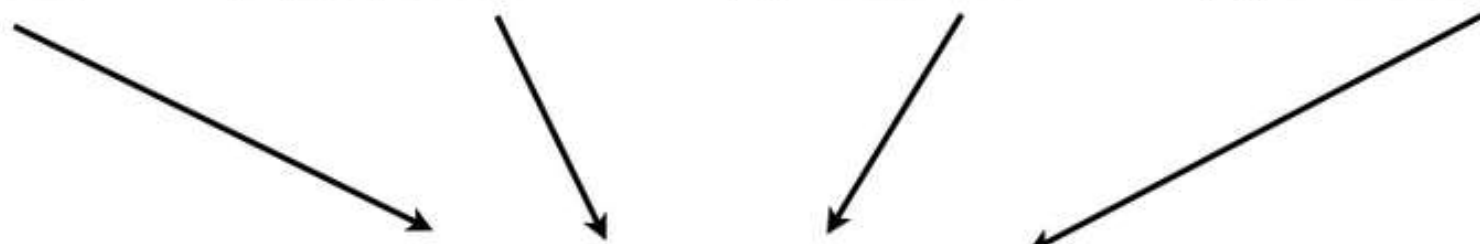
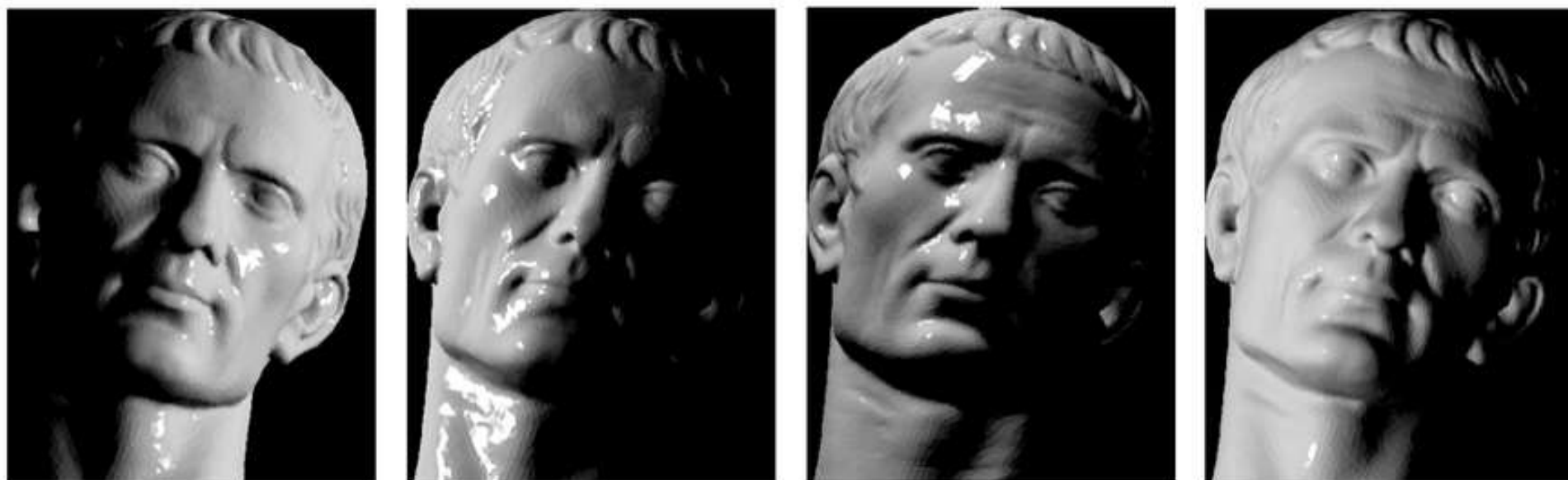
Uffizi
Gallery



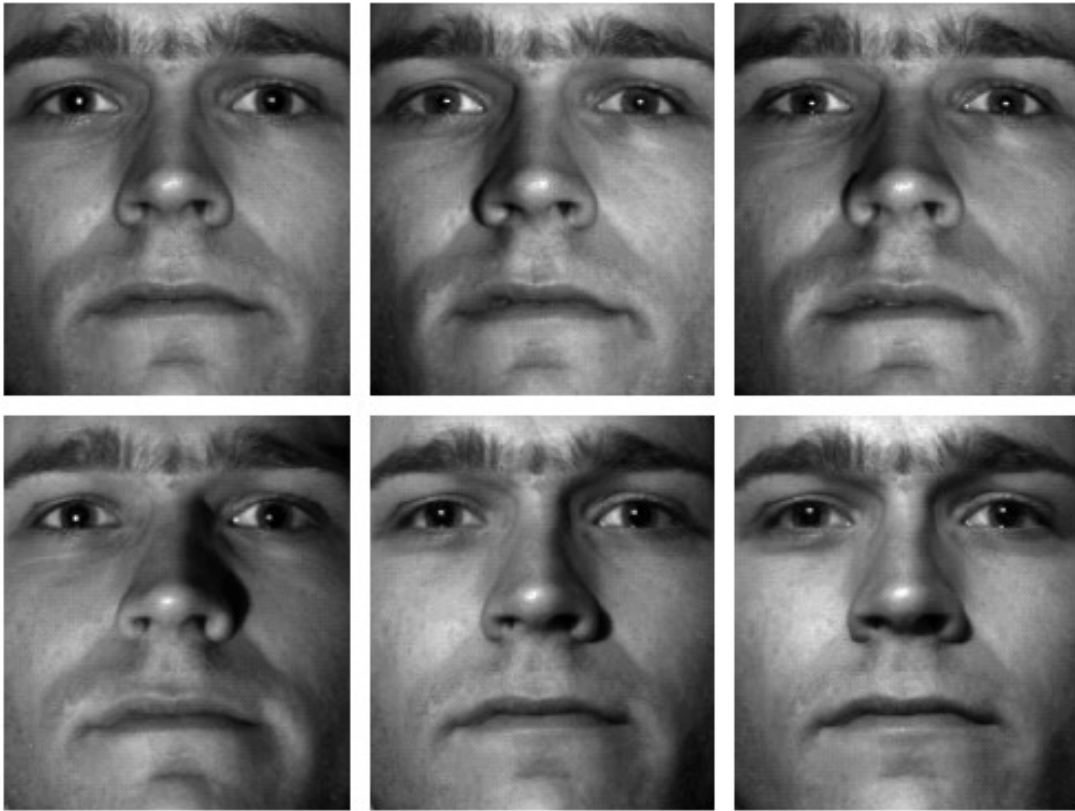
Grace
Cathedral



Lighting Environments from the Light Probe Image Gallery:
<http://www.debevec.org/Probes/>



Results



from Athos Georghiades

Photometric Stereo: Challenges

- Lambertian:

$$I = k_d \mathbf{N} \cdot \mathbf{L}$$
$$\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix} = k_d \mathbf{N}^T \begin{bmatrix} \mathbf{L}_1 & \mathbf{L}_2 & \mathbf{L}_3 \end{bmatrix}$$

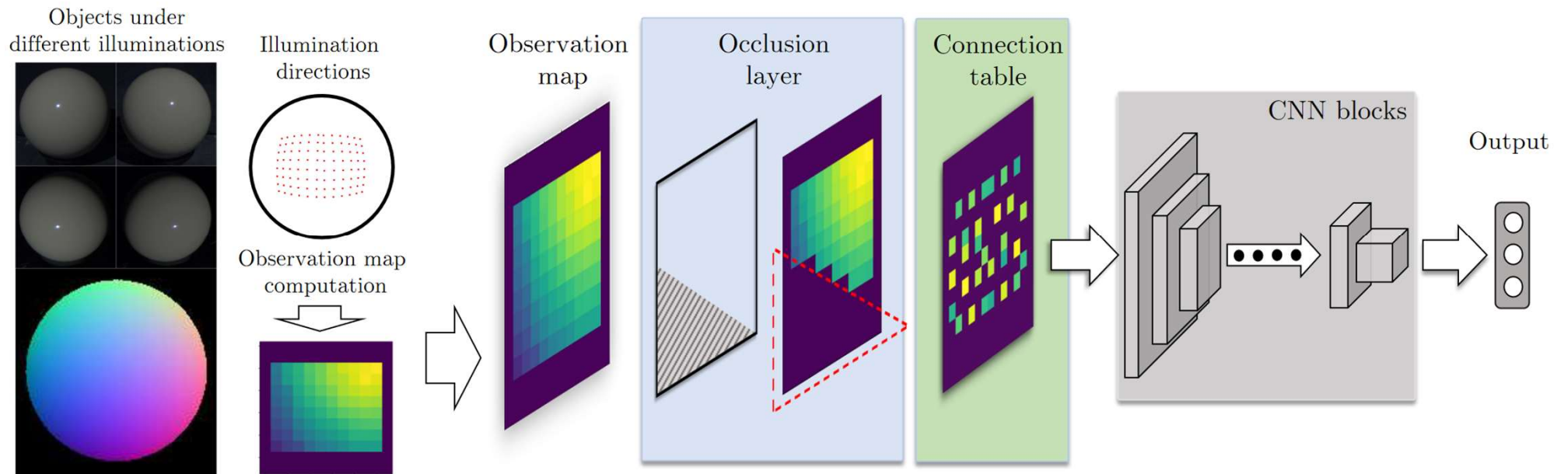
- Directional diffuse



Good luck with that :P

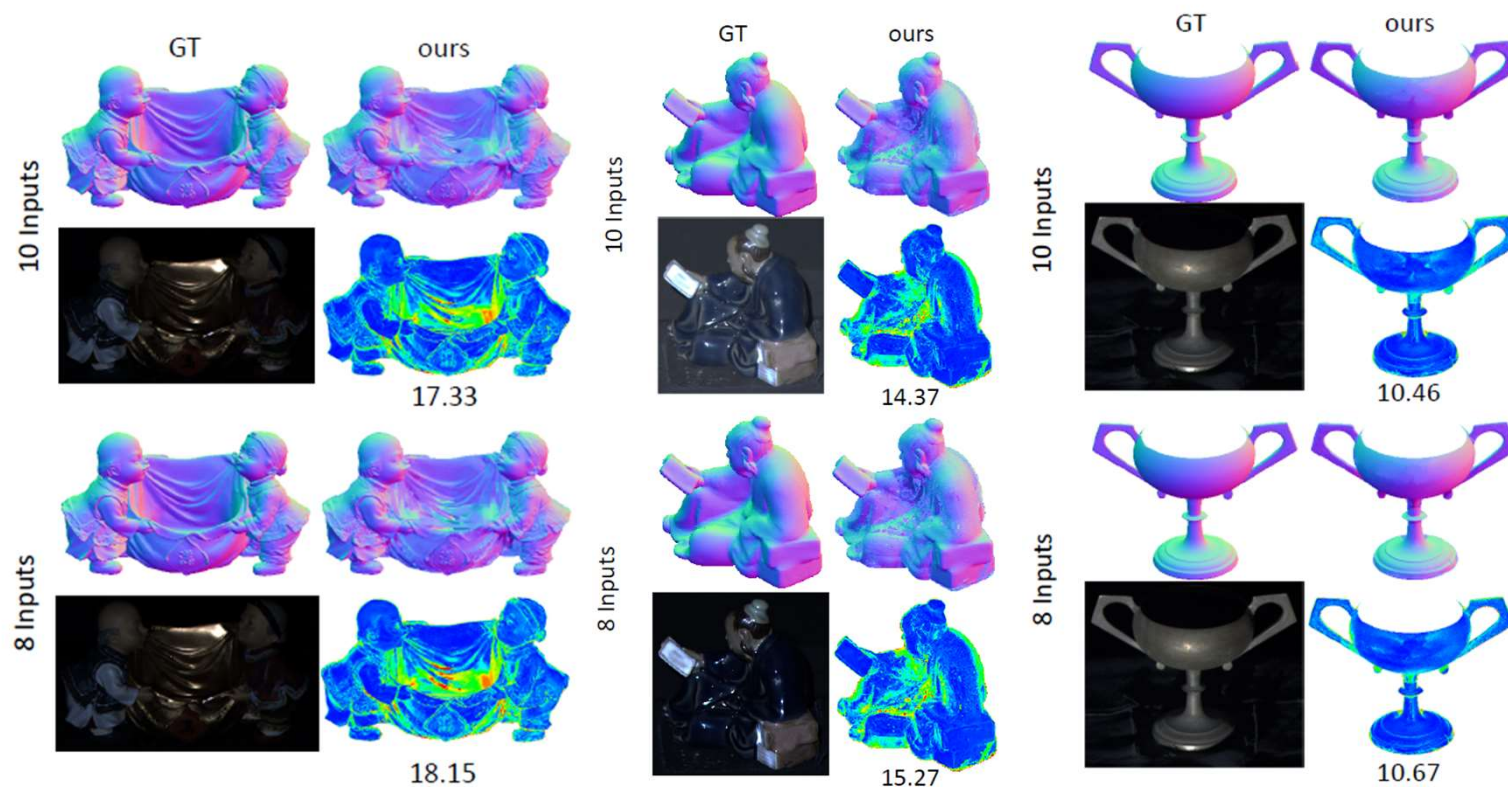
Learning to Minify Photometric Stereo

Junxuan Li, Antonio Robles-Kelly, Shaodi You, Yasuyuki Matsushita
CVPR2019



Learning to Minify Photometric Stereo

Junxuan Li, Antonio Robles-Kelly, Shaodi You, Yasuyuki Matsushita, CVPR2019



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Intrinsic Image Decomposition

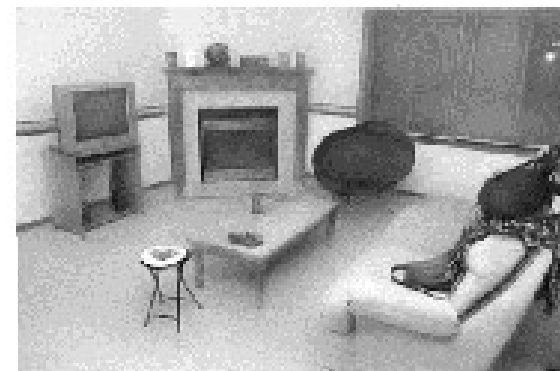
$$I = k_d \mathbf{N} \cdot \mathbf{L}$$



Natural image



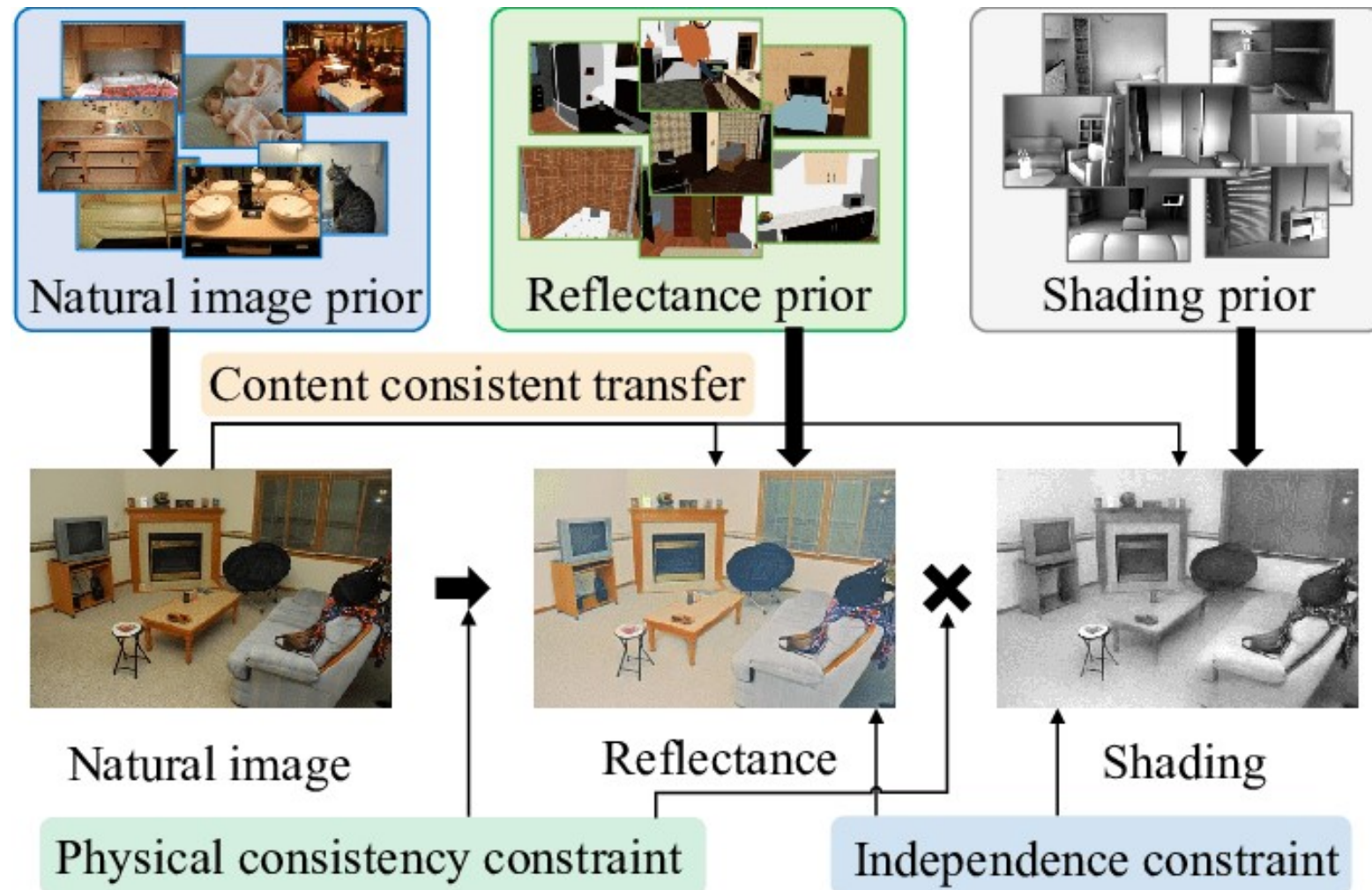
Reflectance



Shading

Unsupervised Learning for Intrinsic Image Decomposition from a Single Image

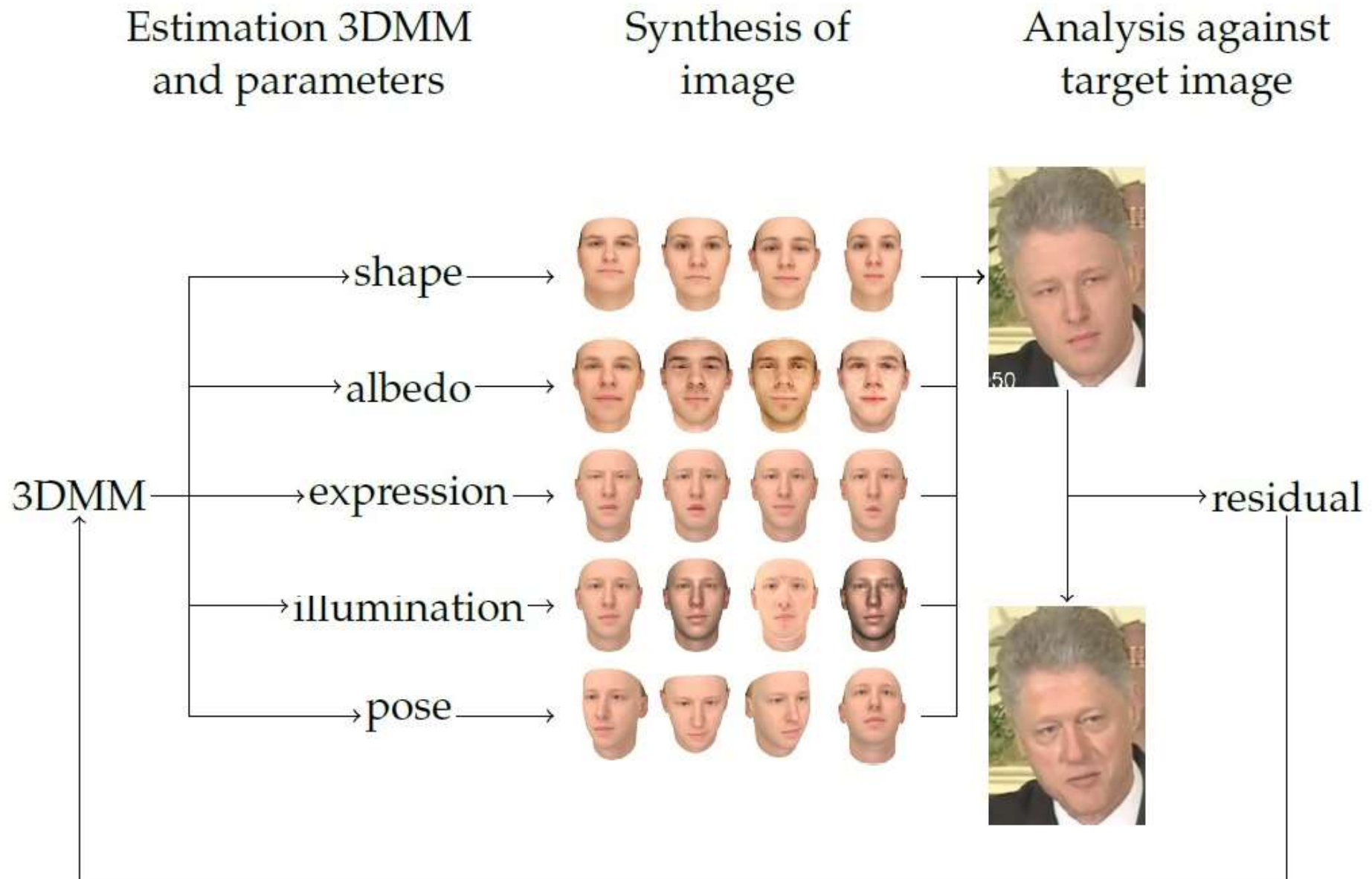
Yunfei Liu, Yu Li, Shaodi You and Feng Lu, CVPR2020

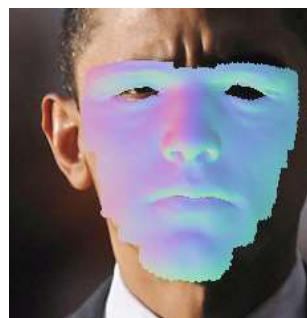
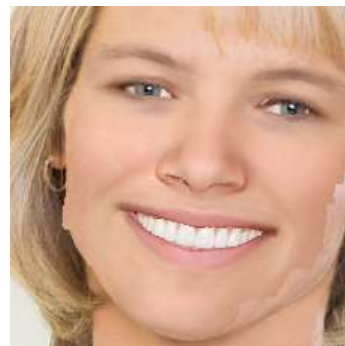


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 - Applications

Face Analysis





Total Relighting: Learning to Relight Portraits for Background Replacement

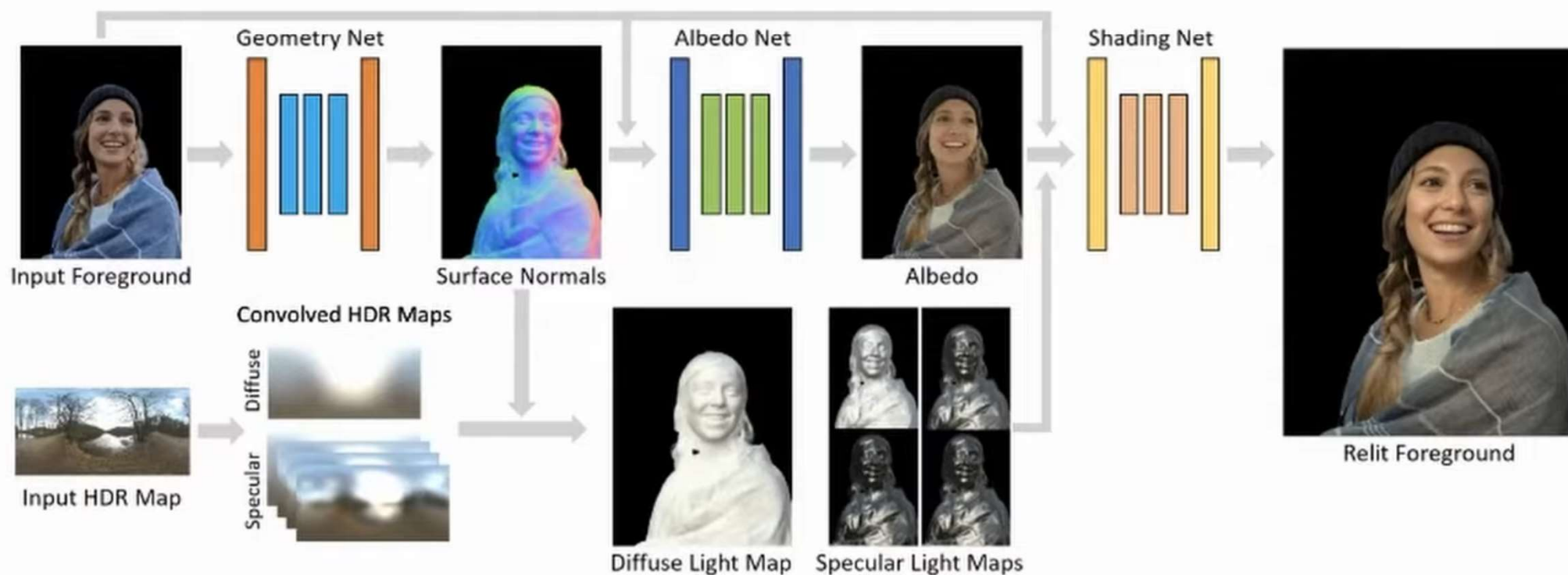


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Deep Relighting Module - Design

Several CNNs with U-Net architecture in series



Action Points

- Working on the first assignments with you group mates, attend lab sessions (laptop college) if you need help from TAs. Feel free to ask questions.
- Work on the exercise (1%), attend the seminar and work on more exercises (not mandatory but encouraged).
- Read textbook and the slides.
- Next lecture on Thursday: neighborhood processing and image convolution.

Disclaimer

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